HAIDE COLLEGE, OCEAN UNIVERSITY OF CHINA

STATISTICAL MODELLING AND INFERENCE

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These notes cover Weeks 1–14 of Statistical Modelling and Inference: statistical inference, confidence intervals, hypothesis testing, two-sample inference, proportions, simple and multiple linear regression, polynomial regression, parameter estimation in simple linear regression, one-way and two-way ANOVA, maximum likelihood estimation, and Fisher information.

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§1.1 *Introduction to Statistical Inference*

Statistical inference is the act of generalising from a **sample** to a **population** with a calculated degree of certainty. A sample is the part of the population we actually observe; the population is the complete collection of objects, people, measurements, or experimental units about which we want to draw a conclusion.

Quantity	Population	Sample
Description	parameter	statistic
Calculated from observed data?	no	yes
Constant for the study?	yes	no, varies from sample to sample
Mean	μ	$\bar{Y}$
Variance	σ^2	S^2
Standard deviation	σ	S

Definition 1.1: Point Estimate

A **point estimate** is a single value calculated from sample data and used as the best available guess for an unknown population parameter. For example, $\bar{Y}$ estimates μ, S^2 estimates σ^2, and S estimates σ.

Definition 1.2: Bias and Mean Squared Error

Let $\hat{\theta}$ be an estimator of a parameter θ. The **bias** of $\hat{\theta}$ is

$$B(\hat{\theta}) = E(\hat{\theta}) - \theta.$$

The **mean squared error** (MSE) is

$$\mathrm{MSE}(\hat{\theta}) = E\left[(\hat{\theta} - \theta)^2\right].$$

An estimator is **unbiased** when $B(\hat{\theta}) = 0$, equivalently $E(\hat{\theta}) = \theta$.

Proposition 1.3: Bias–Variance Decomposition

For any estimator $\hat{\theta}$,

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + \left[B(\hat{\theta})\right]^2.$$

Proof Write

$$\hat{\theta} - \theta = \hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta = \hat{\theta} - E(\hat{\theta}) + B(\hat{\theta}).$$

Then

$$\begin{aligned}
\mathrm{MSE}(\hat{\theta}) &= E\left[\left(\hat{\theta} - E(\hat{\theta}) + B(\hat{\theta})\right)^2\right] \\
&= E\left[(\hat{\theta} - E(\hat{\theta}))^2\right] + 2B(\hat{\theta})E\left[\hat{\theta} - E(\hat{\theta})\right] + \left[B(\hat{\theta})\right]^2 \\
&= \mathrm{Var}(\hat{\theta}) + \left[B(\hat{\theta})\right]^2,
\end{aligned}$$

because $E[\hat{\theta} - E(\hat{\theta})] = 0$. □

Remark Useful Expectation and Variance Rules

For constants a, b, c and random variables X, Y:

$$E(c) = c, \qquad E(aX + b) = aE(X) + b, \qquad \mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X),$$

and

$$\mathrm{Var}(aX + bY) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y).$$

If X and Y are independent, the covariance term is zero.

Example 1.4: Constructing an Unbiased Estimator

Suppose $\hat{\theta}$ estimates θ and $E(\hat{\theta}) = a\theta + b$, where $a \neq 0$ and b are known constants. Find the bias and construct an unbiased estimator of θ.

Solution The bias is

$$B(\hat{\theta}) = E(\hat{\theta}) - \theta = a\theta + b - \theta = (a - 1)\theta + b.$$

To remove the bias, solve $E(\hat{\theta}) = a\theta + b$ for θ:

$$\theta = \frac{E(\hat{\theta}) - b}{a}.$$

Therefore define

$$\hat{\theta}^* = \frac{\hat{\theta} - b}{a}.$$

Then

$$E(\hat{\theta}^*) = \frac{E(\hat{\theta}) - b}{a} = \frac{a\theta + b - b}{a} = \theta,$$

so $\hat{\theta}^*$ is unbiased. □

Example 1.5: Combining Two Unbiased Estimators

Suppose $E(\hat{\theta}_1) = E(\hat{\theta}_2) = \theta$, $\mathrm{Var}(\hat{\theta}_1) = \sigma_1^2$, and $\mathrm{Var}(\hat{\theta}_2) = \sigma_2^2$. Define

$$\hat{\theta}_3 = a\hat{\theta}_1 + (1-a)\hat{\theta}_2.$$

(a) Show that $\hat{\theta}_3$ is an unbiased estimator of θ.

(b) If $\hat{\theta}_1$ and $\hat{\theta}_2$ are independent, choose a to minimise $\mathrm{Var}(\hat{\theta}_3)$.

(c) If the estimators are not independent but $\mathrm{Cov}(\hat{\theta}_1, \hat{\theta}_2) = c \neq 0$, choose a to minimise $\mathrm{Var}(\hat{\theta}_3)$.

Solution **(a) Unbiasedness.**

$$E(\hat{\theta}_3) = a\theta + (1-a)\theta = \theta,$$

so every value of a gives an unbiased estimator.

(b) Independent estimators. When the two estimators are independent,

$$\mathrm{Var}(\hat{\theta}_3) = a^2\sigma_1^2 + (1-a)^2\sigma_2^2.$$

Differentiate with respect to a:

$$\frac{d}{da}\mathrm{Var}(\hat{\theta}_3) = 2a\sigma_1^2 - 2(1-a)\sigma_2^2.$$

Set this equal to zero:

$$a\sigma_1^2 = (1-a)\sigma_2^2 \quad \Longrightarrow \quad a = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}.$$

The estimator with the smaller variance receives the larger weight.

(c) Correlated estimators. When $\mathrm{Cov}(\hat{\theta}_1, \hat{\theta}_2) = c$,

$$\mathrm{Var}(\hat{\theta}_3) = a^2\sigma_1^2 + (1-a)^2\sigma_2^2 + 2a(1-a)c.$$

Expand and differentiate:

$$\frac{d}{da}\mathrm{Var}(\hat{\theta}_3) = 2a\sigma_1^2 - 2(1-a)\sigma_2^2 + 2(1-2a)c.$$

Set the derivative equal to zero:

$$a\sigma_1^2 + a\sigma_2^2 - 2ac = \sigma_2^2 - c \quad \Longrightarrow \quad a = \frac{\sigma_2^2 - c}{\sigma_1^2 + \sigma_2^2 - 2c}.$$

This reduces to the independent-case formula when $c = 0$. □

§1.2 *Key Terminology*

Definition 1.6: Parameter vs. Statistic

A **parameter** is a numeric description of a population (e.g. population mean μ, population variance σ^2). A **statistic** is a numeric description computed from sample data (e.g. sample mean $\bar{Y}$, sample variance S^2).

Definition 1.7: Statistical Inference

Using observed sample statistics to make statements about unknown population parameters. The two main branches are:

- **Estimation** – predicting the parameter by a point estimate (single number) or an interval estimate (range of values).
- **Hypothesis Testing** – deciding whether sample data are consistent with a specified parameter value.

§1.3 *Biased vs. Unbiased Estimators*

Theorem 1.8: Biased and Unbiased Sample Variance

Let $Y_1, Y_2, \ldots, Y_n$ be a random sample with $E(Y_i) = \mu$ and $\text{Var}(Y_i) = \sigma^2$. Define:

$$S'^2 = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{Y})^2 \qquad \text{and} \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2.$$

Then:

(i) S'^2 is a **biased** estimator: $E(S'^2) = \frac{n-1}{n}\sigma^2 \neq \sigma^2$.

(ii) S^2 is an **unbiased** estimator: $E(S^2) = \sigma^2$.

Remark

The divisor $n - 1$ in S^2 corrects for the one degree of freedom "used up" in estimating μ by $\bar{Y}$. In practice, S^2 is always preferred for inference.

§1.4 Confidence Interval for μ – Large Sample

1.4.1 Sampling Distribution

By the Central Limit Theorem, for large n:

$$\bar{Y} \sim N\left(\mu,\ \frac{\sigma}{\sqrt{n}}\right).$$

Since a normal random variable falls within 1.96 standard deviations of its mean with probability 0.95:

$$P\left(\mu - 1.96\,\frac{\sigma}{\sqrt{n}}\ \le\ \bar{Y}\ \le\ \mu + 1.96\,\frac{\sigma}{\sqrt{n}}\right) \approx 0.95.$$

1.4.2 General Formula

A $(1-\alpha)\times 100\%$ confidence interval for μ is:

$$\boxed{\bar{y} \pm z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}}}$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ critical value of the standard normal distribution.

Definition 1.9: Margin of Error

The **margin of error** at the $(1 - \alpha) \times 100\%$ confidence level is $z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}}$. The resulting interval is $\bar{y} \pm$ margin of error.

1.4.3 Common Critical Values

$1-\alpha$	$\alpha/2$	$1-\alpha/2$	$z_{\alpha/2}$
0.90	0.050	0.950	1.645
0.95	0.025	0.975	1.960
0.99	0.005	0.995	2.576

Example 1.10: 90% CI – Supermarket Shopping Time

The shopping times of $n = 64$ randomly selected customers at a supermarket were recorded. The sample mean shopping time was $\bar{y} = 33$ minutes. Suppose that the variance of shopping time is known to be $\sigma^2 = 256$ minutes^2 (so $\sigma = 16$ minutes). Find the 90% two-sided confidence interval for the true average shopping time per customer.

Solution

$$33 \pm 1.645\,\frac{16}{\sqrt{64}} = 33 \pm 3.29 \implies (29.71,\ 36.29).$$

We are 90% confident that the true mean shopping time lies between 29.71 and 36.29 minutes. □

Example 1.11: 95% CI – Apple Weight

A market manager wants to estimate the average weight of a specific variety of apple. He takes a random sample of $n = 36$ apples and finds the sample mean weight is $\bar{y} = 150$ grams. The population standard deviation is known to be $\sigma = 12$ grams. Find a 95% confidence interval for the true average apple weight.

Solution

$$150 \pm 1.96\,\frac{12}{\sqrt{36}} = 150 \pm 3.92 \implies (146.08,\ 153.92).$$

□

§1.5 *Confidence Interval for μ – Small Sample ($n < 30$)*

When $n < 30$ and the population is (approximately) normal, σ is unknown, and we replace it by S. The standardised statistic

$$t = \frac{\bar{Y} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

follows a t-distribution with $\nu = n - 1$ degrees of freedom.

The $(1-\alpha)\times 100\%$ CI is:

$$\boxed{\bar{y} \pm t_{\alpha/2,\nu}\,\frac{S}{\sqrt{n}}} \qquad \text{where } \nu = n - 1.$$

Example 1.12: 90% and 98% CI – Body Temperature

What is the normal body temperature for healthy humans? A random sample of $n = 24$ healthy human body temperatures provided by a doctor has sample mean $\bar{y} = 98.25°$F and sample standard deviation $s = 0.73°$F.

(a) Estimate a 90% confidence interval for the average body temperature of healthy people.

(b) Estimate a 98% confidence interval for the average body temperature of healthy people.

Solution **(a) 90% CI:** $t_{0.05,23} = 1.714$

$$98.25 \pm 1.714 \times \frac{0.73}{\sqrt{24}} = 98.25 \pm 0.255 \implies (97.995,\ 98.505).$$

(b) 98% CI: $t_{0.01,23} = 2.500$

$$98.25 \pm 2.500 \times \frac{0.73}{\sqrt{24}} = 98.25 \pm 0.373 \implies (97.877,\ 98.623).$$

$\square$

Remark

A wider interval corresponds to higher confidence; the 98% CI is wider than the 90% CI.

§1.6 *Hypothesis Testing for μ*

1.6.1 *Framework*

Definition 1.13: Elements of a Hypothesis Test

(1) **Null Hypothesis** H_0: the statement being tested; contains an equality and is assumed true until evidence against it is found.

(2) **Alternative Hypothesis** H_a: contradicts H_0; accepted only if strong evidence from the data supports it. May be one-sided or two-sided.

(3) **Test Statistic**: measures the discrepancy between the observed sample statistic and the value hypothesised under H_0.

(4) **Rejection Region**: the set of test-statistic values that lead to rejecting H_0.

(5) **P-value**: the probability, assuming H_0 is true, of observing data at least as extreme as the actual data in the direction of H_a.

Definition 1.14: One-Tailed and Two-Tailed Tests

A **one-tailed test** is a test in which the alternative hypothesis specifies one direction only:

$$H_a : \theta > \theta_0 \qquad \text{or} \qquad H_a : \theta < \theta_0.$$

The rejection region lies in one tail of the sampling distribution. A **two-tailed test** is a test in which the alternative hypothesis states that the parameter is not equal to the null value:

$$H_a : \theta \neq \theta_0.$$

It simultaneously tests for values greater than and less than the null value, so the rejection probability α is split across the two tails.

Definition 1.15: Type I and Type II Errors

- **Type I Error**: rejecting H_0 when H_0 is true. $P(\text{Type I}) = \alpha$.
- **Type II Error**: failing to reject H_0 when H_a is true. $P(\text{Type II}) = \beta$.

1.6.2 *Test Statistic (Large Sample)*

When testing $H_0 : \mu = \mu_0$ with known (or estimated large-sample) σ:

$$\boxed{Z_{\text{obs}} = \frac{\bar{y} - \mu_0}{\sigma/\sqrt{n}}}$$

Under H_0, $Z_{\text{obs}} \sim N(0, 1)$ approximately.

The numerator $\bar{y} - \mu_0$ is the observed discrepancy between the sample mean and the null value. The denominator $\sigma/\sqrt{n}$ is the standard error of $\bar{Y}$ under repeated sampling. Thus the statistic measures how many standard errors the observed mean lies away from the null hypothesis.

Remark

In practice, σ is unknown and is replaced by s for large samples; this approximation is acceptable when $n \geq 30$.

1.6.3 *Decision Rules*

After the sample statistic has been converted to a Z or t value, the decision can be made either by a rejection region or by a p-value:

- In a rejection-region approach, compare the observed statistic with a critical value determined by α.

- In a p-value approach, compute the probability, under H_0, of observing a statistic at least as extreme as the observed one in the direction of H_a. Reject H_0 when p-value $\leq \alpha$.

Alternative	**Reject H_0 if**	**P-value**
Upper tail: $H_a : \mu > \mu_0$	$Z_{\text{obs}} > z_\alpha$	$P(Z \geq Z_{\text{obs}})$
Lower tail: $H_a : \mu < \mu_0$	$Z_{\text{obs}} < -z_\alpha$	$P(Z \leq Z_{\text{obs}})$
Two-tailed: $H_a : \mu \neq \mu_0$	$\lvert Z_{\text{obs}}\rvert > z_{\alpha/2}$	$2P(Z \geq \lvert Z_{\text{obs}}\rvert)$

Example 1.16: Two-Sided Test – Resident Weight

The average weight of all residents in a town has historically been reported as 168 lbs. A researcher believes that the true mean is different. She measures a random sample of 65 individuals and obtains $\bar{y} = 169.5$ lbs with sample standard deviation $s = 3.9$ lbs. At the 5% significance level, test whether the population mean differs from 168 lbs.

Solution **Step 1 – hypotheses.**

$$H_0 : \mu = 168, \qquad H_a : \mu \neq 168.$$

This is a two-sided large-sample test because the researcher is looking for any difference from 168 lbs and $n = 65$.

Step 2 – test statistic.

$$Z_{\text{obs}} = \frac{169.5 - 168}{3.9/\sqrt{65}} = \frac{1.5}{0.484} \approx 3.10.$$

Step 3 – rejection-region decision. For $\alpha = 0.05$, the two-sided critical values are $\pm z_{0.025} = \pm 1.96$. Since $|3.10| > 1.96$, reject H_0.

Step 4 – p-value decision.

$$\text{p-value} = 2P(Z \geq |3.10|) = 2(1 - 0.9990) = 0.0019.$$

Because $0.0019 < 0.05$, the p-value approach reaches the same decision. There is sufficient evidence that the true mean weight differs from 168 lbs. □

Example 1.17: Lower-Tail Test – Car Warranty

A factory manufactures cars with a warranty of 5 years on the engine and transmission. An engineer believes that the engine or transmission will malfunction in less than 5 years. He tests a sample of 50 cars and finds the average malfunction time to be $\bar{y} = 4.8$ years with standard deviation $s = 0.5$ years. At the 1% significance level, test whether there is enough evidence to revise the warranty.

Solution **Step 1 – hypotheses.**

$$H_0 : \mu = 5, \qquad H_a : \mu < 5.$$

This is a lower-tail test because the engineering claim is "less than 5 years".

Step 2 – test statistic.

$$Z_{\text{obs}} = \frac{4.8 - 5}{0.5/\sqrt{50}} = \frac{-0.2}{0.0707} \approx -2.83.$$

Step 3 – rejection-region decision. For a lower-tail test at $\alpha = 0.01$, reject when $Z_{\text{obs}} < -z_{0.01} = -2.326$. Since $-2.83 < -2.326$, reject H_0.

Step 4 – p-value decision.

$$\text{p-value} = P(Z \leq -2.83) \approx 0.0023.$$

Since $0.0023 < 0.01$, there is sufficient evidence that the mean malfunction time is less than 5 years, so the warranty should be revised. □

§1.7 *t-Test for μ (Small Sample)*

When n is small and σ is unknown, replace Z_{obs} by:

$$\boxed{t_{\text{obs}} = \frac{\bar{y} - \mu_0}{s/\sqrt{n}}} \sim t(n-1).$$

1.7.1 *Decision Rules for t-Test*

Alternative	**Reject H_0 if**	**P-value**
Upper: $\mu > \mu_0$	$t_{\text{obs}} \geq t_{\alpha,\nu}$	$P(t_{(\nu)} \geq t_{\text{obs}})$
Lower: $\mu < \mu_0$	$t_{\text{obs}} \leq -t_{\alpha,\nu}$	$P(t_{(\nu)} \leq t_{\text{obs}})$
Two-sided: $\mu \neq \mu_0$	$\lvert t_{\text{obs}}\rvert \geq t_{\alpha/2,\nu}$	$2P(t_{(\nu)} \geq \lvert t_{\text{obs}}\rvert)$

Example 1.18: Two-Sided t-Test – Flight Time Beijing to Paris

A route from Beijing to Paris has scheduled flight time 580 minutes. A random sample of $n = 31$ flights is selected from all flights on this route. The sample mean flight time is $\bar{y} = 574.1$ minutes and the sample standard deviation is $s = 19.7$ minutes. Test whether the population mean flight time differs from the scheduled time at $\alpha = 0.05$.
Use $H_0 : \mu = 580$ and $H_a : \mu \neq 580$. For the two-sided test with $n - 1 = 30$ degrees of freedom, $t_{0.025,30} = 2.042$.

Solution

$$t_{\text{obs}} = \frac{574.1 - 580}{19.7/\sqrt{31}} = \frac{-5.9}{3.54} \approx -1.67.$$

Critical value (two-sided, $\alpha = 0.05$): $t_{0.025,30} = 2.042$. Since $|-1.67| < 2.042$, do *not* reject H_0.

P-value $= 2P(t_{(30)} \geq 1.67) \approx 0.10 > 0.05$. No significant evidence that the true mean flight time differs from 580 minutes. □

Example 1.19: One-Tailed t-Test – Colour-Word Reading Time

An experiment tests whether the population mean time to read a list of colours is higher when the colour word is written in a different colour. For subject i, let

$$y_i = \text{time for Different colour} - \text{time for Black}.$$

The observed summary is $n = 70$, $\bar{y} = 2.39$, and $s = 7.81$. Test at $\alpha = 0.05$ whether the mean difference is positive.

Solution The hypotheses are

$$H_0 : \mu \leq 0, \qquad H_a : \mu > 0.$$

The large sample makes the normal approximation acceptable, although the same logic can be written as a t test with $\nu = 69$.

$$Z_{\text{obs}} = \frac{2.39 - 0}{7.81/\sqrt{70}} = \frac{2.39}{0.93} = 2.57.$$

For an upper-tail test with $\alpha = 0.05$, the critical value is $z_{0.05} = 1.645$. Since $2.57 > 1.645$, reject H_0. The p-value is

$$P(Z \geq 2.57) = 1 - 0.9949 = 0.0051,$$

also below 0.05. There is evidence that the different-colour condition increases reading time. □

Example 1.20: Two-Sided t-Test – Colour-Word Reading Time

Using the same colour-word data, test whether the mean reading-time difference is affected in either direction when the word is written in a different colour.

Solution The hypotheses are

$$H_0 : \mu = 0, \qquad H_a : \mu \neq 0.$$

The observed statistic is the same:

$$Z_{\text{obs}} = \frac{2.39}{7.81/\sqrt{70}} = 2.57.$$

For $\alpha = 0.05$, reject in a two-sided normal test when $|Z_{\text{obs}}| > z_{0.025} = 1.96$. Since $2.57 > 1.96$, reject H_0. The p-value is

$$2P(Z \geq 2.57) = 2(1 - 0.9949) = 0.0102.$$

The data support a nonzero reading-time effect; because $\bar{y} > 0$, the observed effect is an increase. □

§1.8 *Assignment 1 Practice Problems*

Example 1.21: Assignment 1 Q1 – One Population Mean

A researcher records a random sample from one population and wants to estimate and test the population mean. Use the complete one-sample workflow:

(a) state the parameter and estimator;

(b) construct the appropriate confidence interval;

(c) state H_0 and H_a;

(d) compute the test statistic and p-value;

(e) write the conclusion in the context of the problem.

Solution For any one-sample mean question, the solution chain is:

$$\text{parameter } \mu, \qquad \text{estimator } \bar{Y}, \qquad SE(\bar{Y}) = \begin{cases} \sigma/\sqrt{n}, & \sigma \text{ known,} \\ s/\sqrt{n}, & \sigma \text{ unknown.} \end{cases}$$

Use a normal critical value for large samples or known σ; use a t_{n-1} critical value for small normal samples with unknown σ. The test statistic is

$$Z_{\text{obs}} = \frac{\bar{y} - \mu_0}{\sigma/\sqrt{n}} \quad \text{or} \quad t_{\text{obs}} = \frac{\bar{y} - \mu_0}{s/\sqrt{n}}.$$

The conclusion must compare either the statistic with the critical value or the p-value with α. □

Example 1.22: Assignment 1 Q2 – Students Working While Enrolled

Researchers are concerned about the impact of students working while they are enrolled in classes. A survey of 200 students gives a sample mean of $\bar{y} = 7.10$ hours worked per week, with sample standard deviation $s = 3$ hours.

(i) Construct a 95% confidence interval for the population mean.

(ii) At the 5% significance level, test the hypothesis that students worked more than 7 hours per week. Find the test statistic, rejection region, and conclusion.

Solution **(i) Confidence interval.** Since $n = 200$ is large, use the normal approximation:

$$SE = \frac{s}{\sqrt{n}} = \frac{3}{\sqrt{200}} = 0.2121.$$

$$\bar{y} \pm z_{0.025} SE = 7.10 \pm 1.96(0.2121) = 7.10 \pm 0.416 = (6.684,\ 7.516).$$

(ii) Hypothesis test.

$$H_0 : \mu = 7, \qquad H_a : \mu > 7.$$

$$Z_{\text{obs}} = \frac{7.10 - 7}{3/\sqrt{200}} = \frac{0.10}{0.2121} = 0.47.$$

For an upper-tail test with $\alpha = 0.05$, reject if $Z_{\text{obs}} > z_{0.05} = 1.645$. Since $0.47 < 1.645$, do not reject H_0. The p-value is $P(Z \geq 0.47) \approx 0.319$. The sample does not provide enough evidence that students work more than 7 hours per week on average. □

Example 1.23: Assignment 1 Q3 – Two Population Means

For a two-sample problem, compare two population means using either independent samples or paired samples. The complete answer must identify the sampling design, state the parameter, choose the correct standard error, state hypotheses, compute the statistic, and interpret the conclusion.

Solution For independent samples, the parameter is $\mu_1 - \mu_2$. With large samples,

$$Z_{\text{obs}} = \frac{(\bar{y}_1 - \bar{y}_2) - \Delta_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}.$$

With small normal samples and a common-variance assumption, use

$$S_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}, \qquad t_{\text{obs}} = \frac{(\bar{y}_1 - \bar{y}_2) - \Delta_0}{\sqrt{S_p^2(1/n_1 + 1/n_2)}}.$$

For paired samples, reduce the data to differences d_i and test μ_d using

$$t_{\text{obs}} = \frac{\bar{d} - \Delta_0}{s_d/\sqrt{n}}.$$

In all cases the final conclusion must refer to the original two groups or paired measurements, not only to symbols. □

§1.9 *Power of a Test*

Definition 1.24: Power

The **power** of a test is the probability that the test correctly rejects H_0 when H_0 is false:

$$\text{Power} = P(\text{Reject } H_0 \mid H_0 \text{ false}) = 1 - \beta.$$

When H_0 is true, Power $= P(\text{Type I error}) = \alpha$.

Proposition 1.25: Factors Affecting Power

All else being equal:

(i) Power **increases** as sample size n increases.

(ii) Power **increases** as population variance σ^2 decreases.

(iii) Power **increases** as the true mean moves further from μ_0.

Example 1.26: Power Calculation

Use the mean-difference setting from the interference-effect example. We test whether the population mean interference effect is positive:

$$H_0 : \mu = 0, \qquad H_a : \mu > 0.$$

Assume $\sigma^2 = 64$ and $n = 16$, so the standard error is $\sigma/\sqrt{n} = 8/4 = 2$. At $\alpha = 0.05$, the one-sided decision rule is to reject H_0 when $Z_{\text{obs}} \geq 1.645$. Compute the power when the true mean is $\mu = 3.0$.

Solution **Decision rule** (reject H_0 at $\alpha = 0.05$):

$$Z_{\text{obs}} = \frac{\bar{y} - 0}{2} \geq 1.645 \implies \bar{y} \geq 3.29.$$

Suppose the true mean is $\mu = 3.0$:

$$\text{Power} = P(\bar{Y} \geq 3.29 \mid \mu = 3.0) = P\left(Z \geq \frac{3.29 - 3.0}{2.0}\right) = P(Z \geq 0.145) = 1 - 0.5576 = 0.4424.$$

$\beta = 1 - 0.4424 = 0.5576$. The test has only a 44.2% chance of detecting $\mu = 3$ with this sample size. $\square$

2

Inference for Two Population Means

§2.1 *Sampling Designs*

Definition 2.1: Independent vs. Paired Samples

- **Independent (Parallel) Samples**: units in the two groups are completely different; sample sizes n_1 and n_2 need not be equal.
- **Paired (Crossover) Samples**: each subject receives both treatments, or natural pairs are formed; differences $d_i = Y_{1i} - Y_{2i}$ are analysed.

§2.2 *Large-Sample Inference for $\mu_1 - \mu_2$*

2.2.1 *Parameter and Estimator*

Parameter	$\mu_1 - \mu_2$
Estimator	$\bar{Y}_1 - \bar{Y}_2$
Estimated SE	$\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$
Sampling dist.	$\approx N(0, 1)$ if $n_1, n_2 > 20$

2.2.2 *Large-Sample Test Statistic*

Testing $H_0 : \mu_1 - \mu_2 = \Delta_0$ (typically $\Delta_0 = 0$):

$$\boxed{Z_{\text{obs}} = \frac{(\bar{y}_1 - \bar{y}_2) - \Delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}}$$

2.2.3 *Large-Sample Decision Rules*

Alternative	**Reject H_0 if**	**P-value**
$\mu_1 - \mu_2 > \Delta_0$	$Z_{\text{obs}} \geq z_\alpha$	$P(Z \geq Z_{\text{obs}})$
$\mu_1 - \mu_2 < \Delta_0$	$Z_{\text{obs}} \leq -z_\alpha$	$P(Z \leq Z_{\text{obs}})$
$\mu_1 - \mu_2 \neq \Delta_0$	$\lvert Z_{\text{obs}}\rvert \geq z_{\alpha/2}$	$2P(Z \geq \lvert Z_{\text{obs}}\rvert)$

2.2.4 *Large-Sample $(1-\alpha)\times 100\%$ CI*

$$\boxed{(\bar{y}_1 - \bar{y}_2) \pm z_{\alpha/2}\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

Example 2.2: Vitamin C and Common Colds

The outcome is the number of colds during the study period for each student. Group 1 receives a placebo and Group 2 receives ascorbic acid (Vitamin C):

$$\bar{y}_1 = 2.2, \quad s_1 = 0.12, \quad n_1 = 155, \qquad \bar{y}_2 = 1.9, \quad s_2 = 0.10, \quad n_2 = 208.$$

Conduct a two-sided test comparing the group means and construct a 95% confidence interval for $\mu_1 - \mu_2$.

Solution **Test** $H_0 : \mu_1 - \mu_2 = 0$ vs. $H_a : \mu_1 - \mu_2 \neq 0$, $\alpha = 0.05$:

$$\text{SE} = \sqrt{\frac{0.12^2}{155} + \frac{0.10^2}{208}} = 0.0119 \implies Z_{\text{obs}} = \frac{(2.2 - 1.9) - 0}{0.0119} = 25.3.$$

Since $25.3 > z_{0.025} = 1.96$, reject H_0. Vitamin C significantly reduces colds.

95% CI:

$$0.3 \pm 1.96(0.0119) = 0.30 \pm 0.023 \implies (0.277,\ 0.323).$$

□

Remark Two-Sided Group Comparison

In a two-sided comparison of two independent group means, the null hypothesis is usually

$$H_0 : \mu_1 - \mu_2 = 0,$$

which says that there is no difference in treatment effects. The alternative is

$$H_a : \mu_1 - \mu_2 \neq 0.$$

The sign of $\bar{y}_1 - \bar{y}_2$ tells the observed direction, but the rejection rule uses $|Z_{\text{obs}}|$ or $|t_{\text{obs}}|$ because either direction is evidence against equality.

§2.3 Small-Sample Test for $\mu_1 - \mu_2$ (Equal Variances)

When $n_1, n_2 < 30$, populations are normal, and $\sigma_1^2 = \sigma_2^2 = \sigma^2$ (common variance assumed):

2.3.1 Pooled Variance Estimator

$$\boxed{S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}$$

2.3.2 Pooled t-Test Statistic

$$\boxed{t_{\text{obs}} = \frac{(\bar{y}_1 - \bar{y}_2) - \Delta_0}{\sqrt{S_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}} \sim t(n_1 + n_2 - 2)$$

2.3.3 Small-Sample $(1 - \alpha) \times 100\%$ CI

$$(\bar{y}_1 - \bar{y}_2) \pm t_{\alpha/2,\,\nu} \sqrt{S_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}, \quad \nu = n_1 + n_2 - 2.$$

Example 2.3: Scalp Wound Closure – Stapling vs. Suturing

	Stapling	Suturing
n	15	16
$\bar{y}$	96.92	96.31
s	7.51	8.06

$H_0 : \mu_1 - \mu_2 = 0$, $H_a : \mu_1 - \mu_2 \neq 0$, $\alpha = 0.05$.

Solution **Pooled variance:**

$$S_p^2 = \frac{14(7.51)^2 + 15(8.06)^2}{15 + 16 - 2} = \frac{14(56.40) + 15(64.96)}{29} = 60.83.$$

Test statistic:

$$t_{\text{obs}} = \frac{96.92 - 96.31}{\sqrt{60.83(1/15 + 1/16)}} = \frac{0.61}{2.80} = 0.22.$$

Decision: $|0.22| < t_{0.025,29} = 2.045$. Do not reject H_0.

95% CI:

$$0.61 \pm 2.045(2.80) = 0.61 \pm 5.73 \implies (-5.12,\ 6.34).$$

The interval contains zero, consistent with no significant difference. □

§2.4 *Paired Samples (Crossover Design)*

2.4.1 *Setting and Notation*

Each treatment is applied to every subject (preferably in random order). Define:

$$d_i = Y_{1i} - Y_{2i} \quad \text{(difference for subject } i\text{)}.$$

Definition 2.4: Summary Statistics for Differences

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i, \qquad s_d^2 = \frac{\sum_{i=1}^{n}(d_i - \bar{d})^2}{n-1}, \qquad s_d = \sqrt{s_d^2}.$$

2.4.2 *Paired t-Test Statistic*

Testing $H_0 : \mu_d = \Delta_0$:

$$\boxed{t_{\text{obs}} = \frac{\bar{d} - \Delta_0}{s_d/\sqrt{n}}} \sim t(n-1).$$

Common alternatives are

$$H_a : \mu_d > \Delta_0, \qquad H_a : \mu_d < \Delta_0, \qquad H_a : \mu_d \neq \Delta_0.$$

The corresponding p-values are the upper-tail, lower-tail, and two-tail probabilities from a t distribution with $\nu = n - 1$ degrees of freedom:

$$P(t_\nu \geq t_{\text{obs}}), \qquad P(t_\nu \leq t_{\text{obs}}), \qquad 2P(t_\nu \geq |t_{\text{obs}}|).$$

2.4.3 *Confidence Interval for μ_d*

$$\bar{d} \pm t_{\alpha/2,\, n-1}\,\frac{s_d}{\sqrt{n}}.$$

Example 2.5: Student Evaluation in Mathematics

Students take mathematics in the second semester and are evaluated twice. For student i, let d_i be the difference between the two evaluation scores. The summary statistics are $n = 13$,

$\bar{d} = 1.77$, and $s_d = 1.48$. Conduct a two-sided test for differences between the two evaluations at $\alpha = 0.05$ and construct a 95% confidence interval for μ_d.

Solution The hypotheses are

$$H_0 : \mu_d = 0, \qquad H_a : \mu_d \neq 0.$$

$$t_{\text{obs}} = \frac{1.77}{1.48/\sqrt{13}} = \frac{1.77}{0.41} = 4.31.$$

$|4.31| > t_{0.025,12} = 2.179$. Reject H_0: scores differ significantly between the two evaluations.

95% CI:

$$1.77 \pm 2.179(0.41) = 1.77 \pm 0.89 \implies (0.88,\ 2.66).$$

□

Remark

Paired designs are more powerful than independent-sample designs when the two measurements on each subject are positively correlated, because inter-subject variability cancels in d_i.

3
Inference for Population Proportions

§3.1 Review: Statistic vs. Parameter for Proportions

Quantity	Statistic (sample)	Parameter (population)
Mean	$\bar{X}$	μ
Std. dev.	s	σ
Proportion	$\hat{p}$ (or p_0)	p

§3.2 Standard Errors

Proposition 3.1: Standard Error Formulae

$$\text{Quantitative variable:} \quad SE(\bar{Y}) = \frac{S}{\sqrt{n}}$$

$$\text{Qualitative variable:} \quad SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

§3.3 One-Sample Confidence Interval for p

Definition 3.2: Sample Proportion

$$\hat{p} = \frac{X}{n} = \frac{\text{number of successes}}{\text{sample size}}.$$

For large n, by the CLT, $\hat{p} \sim N\left(p,\ \sqrt{p(1-p)/n}\right)$ approximately. A $(1-\alpha)\times 100\%$ CI for p is:

$$\boxed{\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$$

Example 3.3: Allergy Survey

In a survey of $n = 140$ households, 35% had an allergy to house dust. Construct the 95% confidence interval for the population proportion of households with house-dust allergy.

Solution

$$SE = \sqrt{\frac{0.35(0.65)}{140}} = 0.04.$$

95% CI:

$$0.35 \pm 1.96(0.04) = (0.27,\ 0.43), \quad \text{i.e., } 27\% \leq p \leq 43\%.$$

□

§3.4 *One-Sample Hypothesis Test for p*

Testing $H_0 : p = p_0$, the test statistic is always a Z-test:

$$\boxed{Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}}$$

Remark

Note that the denominator uses p_0 (the null value), *not* $\hat{p}$, because we evaluate the sampling distribution under H_0. The critical values are 1.96 ($\alpha = 0.05$) and 2.576 ($\alpha = 0.01$).

Example 3.4: Diabetic Foot Proportion

Survey: $X = 100$ out of $n = 400$ diabetics have diabetic foot; test whether $p = 0.20$ at $\alpha = 0.05$.
$\hat{p} = 100/400 = 0.25$. $H_0 : p = 0.20$, $H_a : p \neq 0.20$ (two-tailed).

Solution

$$Z = \frac{0.25 - 0.20}{\sqrt{0.20(0.80)/400}} = \frac{0.05}{0.02} = 2.5.$$

$|2.5| > 1.96$. Reject H_0. There is sufficient evidence that the proportion is not 20%. □

§3.5 *Two-Sample Confidence Interval for $p_1 - p_2$*

From two independent samples with $\hat{p}_1 = x_1/n_1$ and $\hat{p}_2 = x_2/n_2$, a $(1 - \alpha) \times 100\%$ CI for $p_1 - p_2$ is:

$$\boxed{(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}}$$

Example 3.5: Teaching Methods A and B

A university wants to compare the effectiveness of two teaching methods A and B. Method A is online learning and Method B is traditional classroom learning. Based on sample data, construct a 95% confidence interval for the difference between the two teaching methods.
Method A: $n_1 = 100$, $x_1 = 85$, $\hat{p}_1 = 0.85$.
Method B: $n_2 = 125$, $x_2 = 75$, $\hat{p}_2 = 0.60$.

Solution

$$SE = \sqrt{\frac{0.85(0.15)}{100} + \frac{0.60(0.40)}{125}} = \sqrt{0.001275 + 0.001920} = \sqrt{0.003195} \approx 0.0565.$$

95% CI:

$$(0.85 - 0.60) \pm 1.96(0.0565) = 0.25 \pm 0.111 \implies (0.139,\ 0.361).$$

Since the interval does not include zero, Method A has a significantly higher pass rate. □

§3.6 *Two-Sample Hypothesis Test for $p_1 - p_2$*

3.6.1 *Hypotheses*

Type	**Hypotheses**
Two-tailed	$H_0 : p_1 = p_2$; $H_a : p_1 \neq p_2$
Right-tailed	$H_0 : p_1 \leq p_2$; $H_a : p_1 > p_2$
Left-tailed	$H_0 : p_1 \geq p_2$; $H_a : p_1 < p_2$

3.6.2 *Pooled Proportion and Test Statistic*

Under $H_0 : p_1 = p_2$, pool both samples to estimate the common proportion:

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}.$$

The test statistic is:

$$\boxed{Z_{\text{obs}} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}}$$

The decision rule is identical to the standard Z-test; reject H_0 if the p-value $< \alpha$.

Example 3.6: Smokers vs. Non-Smokers – Cigarette Tax Opinion

A telephone poll of 800 Chinese adults on taxing cigarettes for health care:

	Non-Smokers	**Smokers**
n	605	195
x (said yes)	351	41
$\hat{p}$	0.58	0.21

$H_0 : p_1 = p_2$, $H_a : p_1 \neq p_2$, $\alpha = 0.05$.

Solution **Pooled proportion:**

$$\hat{p} = \frac{351 + 41}{605 + 195} = \frac{392}{800} = 0.49.$$

Test statistic:

$$Z_{\text{obs}} = \frac{0.58 - 0.21}{\sqrt{0.49(0.51)(1/605 + 1/195)}} = \frac{0.37}{\sqrt{0.49(0.51)(0.00677)}} = \frac{0.37}{0.0418} \approx 8.85.$$

Since $8.85 \gg 1.96$, reject H_0. Non-smokers and smokers differ significantly in their opinion on the cigarette tax. □

4
Simple and Multiple Linear Regression

§4.1 Introduction to Regression Analysis

Definition 4.1: Regression Analysis

A statistical method used to:

- Explain the impact of changes in an **independent variable** on a **dependent variable**.
- Predict the value of a dependent variable based on the value of at least one independent variable.

Dependent variable: the variable we wish to study.
Independent variable: the variable used to study the dependent variable.

Definition 4.2: Simple Linear Regression Model

Only one independent variable, X. The relationship between X and Y is described by a linear function, and changes in Y are assumed to be caused by changes in X.

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

where β_0 is the intercept, β_1 is the slope, and ε is the error term.

Definition 4.3: Population Linear Regression

The **population regression model** is

$$Y = \beta_0 + \beta_1 X + \varepsilon.$$

The deterministic part $\beta_0 \quad + \quad \beta_1 X$ is the population regression line, while ε is the random error at a given value of X. The conditional mean is

$$E(Y \mid X = x) = \beta_0 + \beta_1 x.$$

Definition 4.4: Estimated Regression Model

The sample regression line estimates the population regression line:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i.$$

The fitted value $\hat{Y}_i$ is the predicted response at X_i, and the residual is

$$e_i = Y_i - \hat{Y}_i.$$

§4.2 *Mathematical and Statistical Relations*

A purely mathematical relation has all points exactly on a line, such as $y = a + bx$. A statistical relation allows random error around the line:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i.$$

The scatterplot is therefore essential: it shows whether a straight-line model is reasonable and whether the residual spread looks roughly constant.

§4.3 *Linear Regression Assumptions*

The following assumptions are made about the error values ε:

I. The error values are statistically independent.

II. The error values are normally distributed for any given value of X.

III. The probability distribution of the errors is normal.

IV. The probability distribution of the errors has constant variance (homoscedasticity).

V. The underlying relationship between the X variable and the Y variable is linear.

§4.4 *Least Squares Estimation*

Definition 4.5: Least Squares Criterion

The estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are obtained by minimising the sum of the squared residuals (errors):

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^{n} (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2.$$

Proposition 4.6: The Least Squares Estimates

The formulae for the slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$ are:

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n}}{\sum x_i^2 - \frac{(\sum x_i)^2}{n}},$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Example 4.7: Income and Education

An educational economist wants to establish the relationship between an individual's income and education. A random sample of 10 individuals gives income in thousands of dollars and education in years:

Income y	25	33	22	41	18	28	32	24	53	26
Education x	11	12	11	15	8	10	11	12	17	11

Find the least-squares regression line and interpret the coefficients.

Solution Here income is the dependent variable and education is the independent variable. The necessary totals are

$$\sum x_i = 118, \quad \sum y_i = 302, \quad \sum x_i^2 = 1450, \quad \sum y_i^2 = 10072, \quad \sum x_i y_i = 3779.$$

Thus

$$S_{xy} = \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n} = 3779 - \frac{118(302)}{10} = 215.4,$$

$$S_{xx} = \sum x_i^2 - \frac{(\sum x_i)^2}{n} = 1450 - \frac{118^2}{10} = 57.6.$$

The slope and intercept are

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{215.4}{57.6} = 3.74,$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{302}{10} - 3.74\frac{118}{10} = -13.93.$$

Therefore

$$\hat{y} = -13.93 + 3.74x.$$

The slope means that each additional year of education is associated with an average increase of about \$3.74 thousand in income. The intercept is the predicted income for zero years of education; in this context it is mainly a mathematical anchor and is not substantively meaningful. □

4.4.1 *Error Variance and Standard Error of Estimate*

For simple linear regression,

$$SSE = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 = S_{yy} - \frac{S_{xy}^2}{S_{xx}},$$

and the estimated error variance is

$$s_\varepsilon^2 = MSE = \frac{SSE}{n-2}.$$

For the income–education data,

$$S_{yy} = 10072 - \frac{302^2}{10} = 951.6,$$

$$SSE = 951.6 - \frac{(215.4)^2}{57.6} = 146.09,$$

so

$$s_\varepsilon^2 = \frac{146.09}{10-2} = 18.26, \qquad s_\varepsilon = \sqrt{18.26} = 4.27.$$

The value s_ε is interpreted in the units of the response variable: the observed incomes typically differ from the fitted line by about \$4.27 thousand.

Proposition 4.8: Least Squares Regression Properties

- The sum of the residuals from the least squares regression line is 0: $\sum_{i=1}^{n} (Y_i - \hat{Y}_i) = 0$.
- The sum of the squared residuals is minimised.
- The simple regression line always passes through the point $(\bar{x}, \bar{y})$.
- The least squares coefficients are unbiased estimators of β_0 and β_1.

§4.5 *Residual Analysis*

Definition 4.9: Purposes of Residual Analysis

- Examine the linearity assumption.
- Examine constant variance across all levels of X (homoscedasticity).
- Evaluate the normal-distribution assumption of the errors.

4.5.1 *Graphical Analysis of Residuals*

- Plot residuals vs. X to check for linearity and constant variance.
- Plot a histogram of residuals to check for normality.

§4.6 *Testing the Slope*

To test whether X and Y are linearly related, test

$$H_0 : \beta_1 = 0.$$

For a two-sided alternative, $H_a : \beta_1 \neq 0$; for a positive linear relation, $H_a : \beta_1 > 0$; for a negative linear relation, $H_a : \beta_1 < 0$. The test statistic is

$$t_{\text{obs}} = \frac{\hat{\beta}_1 - 0}{s_{\hat{\beta}_1}}, \qquad s_{\hat{\beta}_1} = \frac{s_\varepsilon}{\sqrt{S_{xx}}},$$

with $n-2$ degrees of freedom. Rejecting H_0 means the data provide evidence of a linear relationship between the variables.

§4.7 *Prediction and Confidence Intervals in Regression*

For a new predictor value x_g, the point prediction is

$$\hat{y}_g = \hat{\beta}_0 + \hat{\beta}_1 x_g.$$

A confidence interval estimates the mean response $E(Y \mid X = x_g)$:

$$\hat{y}_g \pm t_{\alpha/2,n-2} s_\varepsilon \sqrt{\frac{1}{n} + \frac{(x_g - \bar{x})^2}{S_{xx}}}.$$

A prediction interval estimates one future observed response and is wider:

$$\hat{y}_g \pm t_{\alpha/2,n-2} s_\varepsilon \sqrt{1 + \frac{1}{n} + \frac{(x_g - \bar{x})^2}{S_{xx}}}.$$

Example 4.10: Car Price and Odometer Reading

Car dealers use a red book to determine a car's selling price from important features. One feature is the odometer reading. A regression output gives

$$\widehat{\text{Price}} = 6533.38 - 0.031158\,\text{Odometer}.$$

Interpret the fitted slope and predict the price of a car with 40,000 miles on the odometer.

Solution The slope -0.031158 means that each additional mile on the odometer decreases the predicted selling price by about \$0.031158, or 3.1158 cents, on average. For 40,000 miles,

$$\widehat{\text{Price}} = 6533.38 - 0.031158(40000) = 6533.38 - 1246.32 = 5287.06.$$

Thus the fitted price is about \$5,287. □

Example 4.11: Baseball Batting Average and Winning Percentage

Baseball fans often ask whether teams with higher batting averages have higher winning percentages. For a sample of teams, let x be team batting average and y be winning percentage. Suppose the least-squares calculations give

$$\hat{y} = -0.949 + 5.642x, \qquad s_\varepsilon = 0.0567.$$

Use this model to interpret the slope, test whether higher batting average leads to higher winning percentage at the 5% level, and predict the winning percentage for a team with batting average $x = 0.275$.

Solution The positive slope means that an increase of 0.001 in batting average is associated with an increase of about $5.642(0.001) = 0.005642$ in winning percentage. The slope test is

$$H_0 : \beta_1 = 0, \qquad H_a : \beta_1 > 0.$$

Compute

$$t_{\text{obs}} = \frac{\hat{\beta}_1}{s_{\hat{\beta}_1}}$$

using the regression standard error for the slope and compare with $t_{0.05,n-2}$. If $t_{\text{obs}} > t_{0.05,n-2}$, conclude that higher batting average leads to higher winning percentage. The point prediction at $x = 0.275$ is

$$\hat{y} = -0.949 + 5.642(0.275) = 0.6026.$$

The confidence interval for the mean winning percentage uses the regression confidence-interval formula above; a prediction interval for a single team uses the wider prediction-interval formula. □

§4.8 Scatter Plots and Correlation

Definition 4.12: Scatter Plot / Scatter Diagram

Used to show the relationship between two variables.

Definition 4.13: Correlation Analysis

Used to measure the strength of the (linear) association between two variables. Only the strength is considered; no causal effect is implied.

Definition 4.14: Correlation Coefficient

The **population correlation coefficient** ρ measures the strength of the association between the variables. The **sample correlation coefficient** r is an estimate of ρ.

Proposition 4.15: Features of ρ and r

- Unit free.
- Range between -1 and 1.
- The closer to -1, the stronger the negative linear relationship.
- The closer to $+1$, the stronger the positive linear relationship.
- The closer to 0, the weaker the linear relationship.

Proposition 4.16: Calculating the Correlation Coefficient

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{[\sum(x_i - \bar{x})^2]\,[\sum(y_i - \bar{y})^2]}} = \frac{n\sum x_i y_i - (\sum x_i)(\sum y_i)}{\sqrt{[n\sum x_i^2 - (\sum x_i)^2]\,[n\sum y_i^2 - (\sum y_i)^2]}}.$$

Equivalently,

$$r = \hat{\rho} = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{X_i - \bar{X}}{S_X}\right)\left(\frac{Y_i - \bar{Y}}{S_Y}\right) = \frac{S_{XY}}{S_X S_Y}.$$

Example 4.17: Tree Height and Trunk Circumference

A sample of trees gives the following height and trunk-circumference data:

Tree	1	2	3	4	5	6	7	8
Height y	10.0	13.0	15.0	18.0	20.0	22.0	25.0	28.0
Trunk x	8.0	9.5	12.0	15.0	16.5	18.0	21.5	24.0

Compute the fitted line and correlation, and inspect the scatter plot.

Solution The means are $\bar{x} = 15.5625$ and $\bar{y} = 18.875$. The sums of squares are

$$S_{xx} = 214.21875, \qquad S_{xy} = 234.4375, \qquad S_{yy} = 266.875.$$

Thus

$$\hat{\beta}_1 = \frac{234.4375}{214.21875} = 1.0944, \qquad \hat{\beta}_0 = 18.875 - 1.0944(15.5625) = 1.8449.$$

The fitted line is

$$\hat{y} = 1.8449 + 1.0944x.$$

The correlation is

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{234.4375}{\sqrt{214.21875(266.875)}} = 0.9805.$$

This indicates a very strong positive linear association.

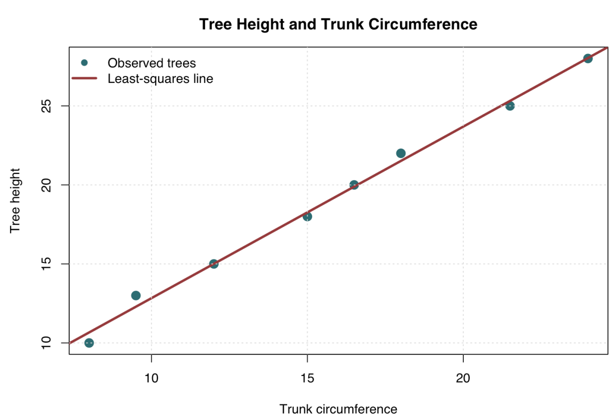

Figure 4.1: Scatter plot of tree height and trunk circumference generated with the R script `scripts/week5_tree_scatter.R`.

□

§4.9 *Coefficient of Determination, R^2*

Definition 4.18: Coefficient of Determination

The coefficient of determination, R^2, is the proportion of the total variation in the dependent variable that is explained by variation in the independent variable(s):

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST},$$

where SSR is the regression sum of squares, SST the total sum of squares, and SSE the sum of squared errors. Note $0 \leq R^2 \leq 1$.

Proposition 4.19: Relationship with Correlation

For a single independent variable, the coefficient of determination is the square of the simple correlation coefficient:

$$R^2 = r^2.$$

§4.10 *Multiple Linear Regression Model*

Definition 4.20: Multiple Linear Regression Model

More than one independent variable, $X_1, X_2, \ldots, X_k$. The relationship between the X_i's and Y is described by a linear function; changes in Y are assumed to be caused by changes in the X_i's.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon.$$

§4.11 *Matrix Approach to Least Squares Estimation*

For a model with k independent variables, write

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where

$$\mathbf{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n\times 1}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \ldots & x_{1k} \\ 1 & x_{21} & x_{22} & \ldots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \ldots & x_{nk} \end{bmatrix}_{n\times(k+1)}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}_{(k+1)\times 1}, \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}_{n\times 1}.$$

Proposition 4.21: Least Squares Estimates (Matrix Form)

The least squares estimate of the parameter vector $\boldsymbol{\beta}$ is

$$\boxed{\widehat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{Y}.}$$

The fitted values are

$$\widehat{\mathbf{Y}} = \mathbf{X}\widehat{\boldsymbol{\beta}}.$$

Example 4.22: House Price, Age, and Area

The price (y, in \$1000 AUD), age (x_1, in decades), and area (x_2, in 1000 m²) of five randomly selected houses are given. Fit a multiple linear regression model.

y	100	80	104	94	130
x_1	1	5	5	10	20
x_2	1	1	2	2	3

Solution Fitting $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ via the matrix approach. Construct the design matrix and response vector

$$\mathbf{X} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 5 & 2 \\ 1 & 10 & 2 \\ 1 & 20 & 3 \end{pmatrix}, \qquad \mathbf{Y} = \begin{pmatrix} 100 \\ 80 \\ 104 \\ 94 \\ 130 \end{pmatrix}.$$

Then

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 5 & 41 & 9 \\ 41 & 551 & 96 \\ 9 & 96 & 19 \end{pmatrix}, \qquad \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 508 \\ 4560 \\ 966 \end{pmatrix}.$$

Solving $\widehat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{Y}$ gives

$$\widehat{\beta}_0 = 66.1252, \qquad \widehat{\beta}_1 = -0.3794, \qquad \widehat{\beta}_2 = 21.4365.$$

The estimated regression model is

$$\hat{y} = 66.1252 - 0.3794\, x_1 + 21.4365\, x_2.$$

$\square$

§4.12 *ANOVA Summary Table for the Regression*

Definition 4.23: ANOVA Summary Table

Used to test the overall significance of the regression model. For the house-price example, testing $H_0 : \beta_1 = \beta_2 = 0$ at the 5% significance level:

Source	df	SS	MSS	F	P-value
x_1	1	724.17	724.17	3.7845	0.1911
x_2	1	232.33	232.33	1.2141	0.3854
Error	2	382.70	191.35		
Total	4				

Conclusion: Since both p-values exceed 0.05, we fail to reject $H_0 : \beta_1 = \beta_2 = 0$.

5
Polynomial Regression and Non-Linear Effects

§5.1 Introduction to Polynomial Regression

Definition 5.1: Polynomial Regression Model

A form of regression analysis in which the relationship between the independent variable X and the dependent variable Y is modelled as an n^{th}-degree polynomial. The model is linear in the parameters:

$$Y = A + BX + CX^2 + DX^3 + \cdots + QX^{N-1}.$$

The highest power $(N-1)$ is the **order** of the polynomial.

Definition 5.2: Order and Shape of Polynomial Regression

- **Linear** ($Y = A + BX$): zero bends.
- **Quadratic** ($Y = A + BX + CX^2$): one bend.
- **Cubic** ($Y = A + BX + CX^2 + DX^3$): two bends.

There is one fewer bend than the highest order in the polynomial model. The sign of the coefficient on the highest-order regressor determines the direction of the curve.

Definition 5.3: Interpreting Polynomial Coefficients

- b_0 interpretation is unchanged: predicted value when all predictors are 0, though its value typically changes with model complexity.

- b_1 is the linear (simple) effect of X at $X = 0$; this is a conditional effect.
- Higher-order terms describe how the linear effect changes across the distribution of X.

§5.2 *Fitting and Comparing Polynomial Models*

Example 5.4: Predicting Interest in Electives

How does the number of electives (X) taken predict interest (Y) in further electives? A linear model is first fit:

$$\texttt{Interest} = 8.7 + 1.4 \times \texttt{Electives}.$$

The linearity assumption is violated, indicating a poor fit. A quadratic model is then fit:

$$\texttt{Interest} = 1.0 + 3.3 \times \texttt{Electives} - 0.1 \times \texttt{Electives}^2.$$

Solution **R code for the linear model:**

```
scatterplot(dE$Electives, dE$Interest, cex=1.5, lwd=2,
            xlab="Electives", ylab="Interest", col="black")
mLinear <- lm(Interest ~ Electives, data=dE)
summary(mLinear)
modelAssumptions(mLinear, Type="linear", one.page=FALSE)
```

The diagnostic plots indicate that the linearity assumption is questionable: the relationship bends rather than staying straight across the range of electives.

R code for the quadratic model:

```
dE$Electives2 <- dE$Electives * dE$Electives
mQuad <- lm(Interest ~ Electives + Electives2, data=dE)

# Equivalent compact syntax:
mQuad <- lm(Interest ~ Electives + I(Electives^2), data=dE)
summary(mQuad)
anova(mLinear, mQuad)
```

Model comparison:

- **Linear model:** $R^2 = 0.5608$, SSE $= 980.2$, error df $= 98$.
- **Quadratic model:** $R^2 = 0.6726$, SSE $= 730.8$, error df $= 97$.

The quadratic model has a higher R^2 and lower SSE. The coefficient on the quadratic term (-0.1266) is significant ($p < 0.001$), confirming the need for the higher-order term.

The nested-model comparison is

$$F = \frac{(SSE_{\text{linear}} - SSE_{\text{quadratic}})/(98-97)}{SSE_{\text{quadratic}}/97} = \frac{(980.2-730.8)/1}{730.8/97} = 33.10.$$

The large F statistic gives strong evidence that adding the quadratic term improves the model. □

Remark

To test whether a polynomial model is necessary (e.g. quadratic vs. linear), compare the augmented model (with the higher-order term) to the compact model (without it). Equivalently, test whether the coefficient on the higher-order term is zero; an F-test or t-test on the specific coefficient may be used.

§5.3 *Conditional Effects and Simple Slopes*

Definition 5.5: Simple Effect (Slope) Formula

The simple slope – the instantaneous rate of change of Y with respect to X at a specific value of X – is obtained by partial differentiation. For a polynomial model $Y = \beta_0 + \beta_1 X + \beta_2 X^2$:

$$\frac{\partial Y}{\partial X} = \beta_1 + 2\beta_2 X.$$

This describes how the effect of X on Y changes across the range of X.

Example 5.6: Simple Slopes for the Electives Model

For the quadratic model Interest $= 1.0 + 3.27 \times$ Electives $- 0.13 \times$ Electives^2.

Solution The simple-slope formula is

$$\frac{\Delta\text{Interest}}{\Delta\text{Electives}} = 3.27 - 0.26 \times \text{Electives}.$$

Evaluated at different points:

- Electives $= 0$: slope $= 3.27 - 0.26(0) = 3.3$.
- Electives $= \bar{X} = 8.17$: slope $= 3.27 - 0.26(8.17) = 1.2$.

The effect of taking an additional elective is strongest for those who have taken few electives and diminishes as the number of electives taken increases. □

5.3.1 Simple Slopes for Interactions

For an interaction model

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ + \varepsilon,$$

the simple effect of X is the partial derivative

$$\frac{\partial Y}{\partial X} = \beta_1 + \beta_3 Z.$$

This means the slope of X depends on the value of Z. If Z is centred, then β_1 is the effect of X when Z is at its mean. For a model that combines quadratic and interaction terms, such as

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 Z + \beta_4 XZ + \varepsilon,$$

the simple effect of X is

$$\frac{\partial Y}{\partial X} = \beta_1 + 2\beta_2 X + \beta_4 Z.$$

§5.4 Multiple Regression with Non-Linear Effects

Definition 5.7: Model with Multiple Predictors and Non-Linearity

Non-linear effects of an individual variable can be evaluated in models containing other variables. For example, examining the effect of Age and weekly Training Miles on 5K race time for cross-country skiers, where a quadratic effect for Miles is expected (diminishing returns).

Example 5.8: 5K Race Times, Age, and Training Miles

The compact model includes only centred Age (cAge). The augmented model adds centred Miles (cMiles) and its square (cMiles^2):

$$\begin{aligned}
\text{Compact:} \quad & \text{Time} = 23.55 + 0.217\,\text{cAge}, \\
& R^2 = 0.2245, \qquad \text{SSE} = 1714.7, \\
\text{Augmented:} \quad & \text{Time} = 22.04 + 0.168\,\text{cAge} - 0.256\,\text{cMiles} \\
& \qquad + 0.00803\,\text{cMiles}^2, \\
& R^2 = 0.7271, \qquad \text{SSE} = 603.4.
\end{aligned}$$

Solution **R code for the compact and augmented models:**

```
mC <- lm(Time ~ cAge, data=dMiles)
summary(mC)
```

```
mA <- lm(Time ~ cAge + cMiles + I(cMiles^2), data=dMiles)
summary(mA)
anova(mC, mA)
```

Overall effect of Miles: Model comparison shows a significant overall effect of miles on 5K times, $F(2, 76) = 69.98$, $p < .001$, with weekly mileage accounting for 64.8% of the unexplained variance after controlling for age ($\Delta R^2 = 0.648$).

Reporting the effects:

- Age effect (controlling for Miles): $b = 0.168$, $\Delta R^2 = 0.132$, $t(76) = 6.05$, $p < .001$.
- Linear Miles effect (at average mileage): $b = -0.256$, $\Delta R^2 = 0.437$, $t(76) = -11.04$, $p < .001$.
- Quadratic Miles effect: $b = 0.00803$, $\Delta R^2 = 0.062$, $t(76) = 4.165$, $p < .001$.

Simple slopes for Miles. The centred model gives the average linear effect of weekly miles at mean mileage:

$$\frac{\partial \text{Time}}{\partial \text{cMiles}} = -0.256 + 2(0.00803)\text{cMiles}.$$

The uncentred display in the slides is

$$\widehat{\text{Time}} = 36.9 + 0.17\,\text{cAge} - 0.74\,\text{Miles} + 0.01\,\text{Miles}^2,$$

so

$$\frac{\partial \text{Time}}{\partial \text{Miles}} = -0.74 + 0.02\,\text{Miles}.$$

The negative linear component means more weekly mileage initially predicts faster 5K times; the positive quadratic component means the improvement diminishes as mileage becomes high. □

Example 5.9: Testing Simple Slopes for Mileage

Using the same 5K data, test the mileage slope for runners with average, low, and high weekly mileage.

Solution The re-centring code used in the slides is:

```
dMiles$hMiles <- dMiles$cMiles - sd(dMiles$cMiles)
mHigh <- lm(Time ~ cAge + hMiles + I(hMiles^2), data=dMiles)
summary(mHigh)

dMiles$lMiles <- dMiles$cMiles + sd(dMiles$cMiles)
mLow <- lm(Time ~ cAge + lMiles + I(lMiles^2), data=dMiles)
summary(mLow)
```

Re-centring changes the point at which the linear coefficient is evaluated. The reported conclusions are:

- At average weekly mileage, one extra mile is associated with a 0.26 minute decrease in 5K time ($p < .001$).
- At low weekly mileage, one extra mile is associated with a 0.48 minute decrease in 5K time ($p < .001$).
- At high weekly mileage, one extra mile is associated with a non-significant 0.03 minute decrease in 5K time ($p = 0.556$).

Thus the benefit of additional training miles is strongest for low-mileage runners and fades at high mileage. □

Remark

Regressors in polynomial regression (e.g. X and X^2) are often highly correlated unless X is centred. Centring X changes the interpretation of the linear coefficient b_1 to be the simple effect of X at the mean of X (i.e. when $X = 0$ in centred form). Without centring, b_1 is the simple effect of X at $X = 0$, which may be unrealistic.

§5.5 *Writing a Results Section for Polynomial Regression*

Example 5.10: Sample Results Section Excerpt

"We regressed Interest on regressors that modelled the linear and quadratic effects of Number of Electives. The overall effect of Number of Electives was significant, $F(2, 97) = 99.62$, $p < .0001$, with Number of Electives accounting for 67.3% of the total variance in Interest. The linear effect of Electives was significant, $B = 1.2$, $\eta_p^2 = .651$, $t(97) = 13.45$, $p < .0001$, indicating that taking an additional elective was associated with a 1.2-point increase in interest for participants who had already taken an average number of electives. However, the quadratic effect of Electives was also significant, $B = -0.1$, $t(97) = 5.75$, $p < .0001$, indicating that the magnitude of the Electives effect decreased by 0.02 for every additional Elective taken."

6

Parameter Estimation in Simple Linear Regression

§6.1 Setup and Notation

Consider the simple linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad i = 1, 2, \ldots, n$$

where $Y_1, Y_2, \ldots, Y_n$ are **independent** with:

$$E(Y_i) = \beta_0 + \beta_1 X_i \qquad \text{and} \qquad \text{Var}(Y_i) = \sigma^2.$$

The OLS estimators are:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X},$$

where:

$$S_{xx} = \sum_{i=1}^{n} (X_i - \bar{X})^2, \quad S_{xy} = \sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y}), \quad S_{yy} = \sum_{i=1}^{n} (Y_i - \bar{Y})^2.$$

§6.2 Unbiasedness and Variance of Estimators

Theorem 6.1: Unbiasedness and Variance of Estimators

Suppose $Y_1, \ldots, Y_n$ are independent with $E(Y_i) = \beta_0 + \beta_1 X_i$ and $\text{Var}(Y_i) = \sigma^2$. Then:

(i) **Unbiasedness:**

$$E(\hat{\beta}_0) = \beta_0 \qquad \text{and} \qquad E(\hat{\beta}_1) = \beta_1.$$

(ii) **Variance of $\hat{\beta}_0$:**

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{S_{xx}} \right).$$

(iii) **Variance of $\hat{\beta}_1$:**

$$\operatorname{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}.$$

(iv) **Covariance:**

$$\operatorname{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{X}}{S_{xx}}.$$

Proof Use

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \sum_{i=1}^{n} a_i Y_i, \qquad a_i = \frac{X_i - \bar{X}}{S_{xx}},$$

because $\sum (X_i - \bar{X})\bar{Y} = 0$. The weights satisfy

$$\sum_{i=1}^{n} a_i = 0, \qquad \sum_{i=1}^{n} a_i X_i = \frac{\sum (X_i - \bar{X}) X_i}{S_{xx}} = \frac{S_{xx}}{S_{xx}} = 1.$$

Therefore

$$E(\hat{\beta}_1) = \sum a_i E(Y_i) = \sum a_i (\beta_0 + \beta_1 X_i) = \beta_0 \sum a_i + \beta_1 \sum a_i X_i = \beta_1.$$

Since the Y_i are independent and all have variance σ^2,

$$\operatorname{Var}(\hat{\beta}_1) = \sum a_i^2 \sigma^2 = \sigma^2 \frac{\sum (X_i - \bar{X})^2}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}}.$$

Next,

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

Thus

$$E(\hat{\beta}_0) = E(\bar{Y}) - \bar{X} E(\hat{\beta}_1).$$

Now

$$E(\bar{Y}) = \frac{1}{n} \sum (\beta_0 + \beta_1 X_i) = \beta_0 + \beta_1 \bar{X},$$

so $E(\hat{\beta}_0) = \beta_0$.

For the variance,

$$\operatorname{Var}(\hat{\beta}_0) = \operatorname{Var}(\bar{Y} - \bar{X}\hat{\beta}_1) = \operatorname{Var}(\bar{Y}) + \bar{X}^2 \operatorname{Var}(\hat{\beta}_1) - 2\bar{X} \operatorname{Cov}(\bar{Y}, \hat{\beta}_1).$$

But

$$\operatorname{Cov}(\bar{Y}, \hat{\beta}_1) = \operatorname{Cov}\left(\frac{1}{n} \sum Y_i, \sum a_i Y_i\right) = \frac{\sigma^2}{n} \sum a_i = 0.$$

Therefore

$$\operatorname{Var}(\hat{\beta}_0) = \frac{\sigma^2}{n} + \bar{X}^2 \frac{\sigma^2}{S_{xx}} = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{S_{xx}}\right).$$

Finally,

$$\operatorname{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \operatorname{Cov}(\bar{Y} - \bar{X}\hat{\beta}_1, \hat{\beta}_1) = 0 - \bar{X} \frac{\sigma^2}{S_{xx}} = -\frac{\sigma^2 \bar{X}}{S_{xx}}.$$

□

Remark

The negative covariance means that if $\hat{\beta}_1$ overestimates the slope, $\hat{\beta}_0$ tends to underestimate the intercept, which makes geometric sense since the fitted line passes through $(\bar{X}, \bar{Y})$.

§6.3 *Properties of Fitted Values*

Theorem 6.2: Properties of Fitted Values

Under the same assumptions, for any given value X_0:

(i) **Unbiasedness of prediction:**

$$E(\hat{\beta}_0 + \hat{\beta}_1 X_0) = \beta_0 + \beta_1 X_0.$$

(ii) **Variance of the fitted value:**

$$\operatorname{Var}(\hat{\beta}_0 + \hat{\beta}_1 X_0) = \sigma^2 \left(\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{S_{xx}} \right).$$

Proof For a fixed X_0, the fitted mean is

$$\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 X_0.$$

Using the unbiasedness just proved,

$$E(\hat{Y}_0) = E(\hat{\beta}_0) + X_0 E(\hat{\beta}_1) = \beta_0 + \beta_1 X_0.$$

For the variance,

$$\begin{aligned}
\operatorname{Var}(\hat{Y}_0) &= \operatorname{Var}(\hat{\beta}_0 + X_0 \hat{\beta}_1) \\
&= \operatorname{Var}(\hat{\beta}_0) + X_0^2 \operatorname{Var}(\hat{\beta}_1) + 2X_0 \operatorname{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\
&= \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{S_{xx}} \right) + X_0^2 \frac{\sigma^2}{S_{xx}} - 2X_0 \frac{\sigma^2 \bar{X}}{S_{xx}} \\
&= \sigma^2 \left(\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{S_{xx}} \right).
\end{aligned}$$

□

Remark

The variance is minimised when $X_0 = \bar{X}$ and increases as X_0 moves away from the centre of the data. This is why prediction intervals widen towards the edges of the observed X-range.

§6.4 *Properties of Residuals*

Define the residuals $\hat{e}_i = Y_i - \hat{Y}_i$, where $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$.

Proposition 6.3: Properties of OLS Residuals

(i) **Zero sum:**

$$\sum_{i=1}^{n} \hat{e}_i = 0.$$

(ii) **Orthogonality to predictors:**

$$\sum_{i=1}^{n} \hat{e}_i X_i = 0.$$

(iii) **Zero mean:**

$$E(\hat{e}_i) = 0.$$

(iv) **Variance of residuals:**

$$\text{Var}(\hat{e}_i) = \sigma^2 \left(1 - \frac{1}{n} - \frac{(X_i - \bar{X})^2}{S_{xx}}\right).$$

Proof The residual sum is

$$\sum_{i=1}^{n} \hat{e}_i = \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = n\bar{Y} - n\hat{\beta}_0 - \hat{\beta}_1 n\bar{X} = 0,$$

because $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$. Similarly,

$$\sum_{i=1}^{n} X_i \hat{e}_i = \sum X_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$$

is the second normal equation from minimising $\sum e_i^2$ with respect to the slope.

Write the fitted value at X_i as $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$. Since $E(\hat{Y}_i) = \beta_0 + \beta_1 X_i = E(Y_i)$, it follows that

$$E(\hat{e}_i) = E(Y_i - \hat{Y}_i) = 0.$$

For the residual variance, use the hat-matrix leverage for simple linear regression,

$$h_{ii} = \frac{1}{n} + \frac{(X_i - \bar{X})^2}{S_{xx}}.$$

The residual is $(1 - h_{ii})Y_i$ minus a linear combination of the remaining observations, and the standard OLS variance result is

$$\text{Var}(\hat{e}_i) = \sigma^2(1 - h_{ii}) = \sigma^2 \left(1 - \frac{1}{n} - \frac{(X_i - \bar{X})^2}{S_{xx}}\right).$$

□

Remark

Properties (i) and (ii) follow directly from the normal equations of OLS. Note that $\mathrm{Var}(\hat{e}_i) < \sigma^2$ because fitting uses up two degrees of freedom (for $\hat{\beta}_0$ and $\hat{\beta}_1$).

§6.5 *Estimation of σ^2*

To estimate the error variance σ^2, we use the **residual (error) mean square**:

$$S_e^2 = \frac{1}{n-2}\left(S_{yy} - \hat{\beta}_1 S_{xy}\right).$$

The numerator is the residual sum of squares:

$$SSE = \sum (Y_i - \hat{Y}_i)^2 = S_{yy} - \hat{\beta}_1 S_{xy} = S_{yy} - \frac{S_{xy}^2}{S_{xx}}.$$

The denominator is $n-2$ because two parameters, β_0 and β_1, have been estimated from the data. Under the linear-model assumptions,

$$\frac{SSE}{\sigma^2} \sim \chi^2_{n-2},$$

so $E(SSE) = (n-2)\sigma^2$ and therefore $E(S_e^2) = \sigma^2$.

Remark

S_e^2 is an unbiased estimator of σ^2, i.e. $E(S_e^2) = \sigma^2$. The denominator $n-2$ reflects the two degrees of freedom lost in estimating $\hat{\beta}_0$ and $\hat{\beta}_1$.

7
One-Way Analysis of Variance (ANOVA)

§7.1 Motivation

One-Way ANOVA tests whether the **means of several groups** (defined by a single categorical factor) are equal. Typical questions:

- Do different market segments differ in product consumption volume?
- Does store familiarity (high/medium/low) affect preference?

§7.2 Hypotheses

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k,$$
$$H_1 : \text{at least one population mean differs.}$$

Remark

H_1 does *not* require all means to be different; only one pair needs to differ.

§7.3 Statistics Associated with One-Way ANOVA

The null hypothesis of equal category means is tested by an **F statistic**, formed as the ratio of the between-group variance to the within-group variance. These variances are derived from the following sums of squares, which use several common alias names:

- SS_{between} (also SS_x): variation in Y explained by the categories of X.
- SS_{within} (also SS_{error}, SSE): variation in Y *not* explained by X.
- SS_y (also SST, TSS): the total variation in Y.

§7.4 *Partitioning of Total Variation*

With k groups each of size n (balanced design), total observations $N = nk$:

Definition 7.1: Sum of Squares

$$\text{SST (Total)} = \sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{\bar{y}})^2,$$

$$\text{SSB (Between groups / Regression)} = n\sum_{i=1}^{k}(\bar{y}_{i\bullet} - \bar{\bar{y}})^2,$$

$$\text{SSW (Within groups / Error)} = \sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i\bullet})^2.$$

The fundamental identity is:

$$\text{SST} = \text{SSB} + \text{SSW},$$

where:

- $\bar{y}_{i\bullet} = \frac{1}{n}\sum_{j=1}^{n} y_{ij}$ is the i-th group mean.
- $\bar{\bar{y}} = \frac{1}{N}\sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}$ is the grand mean.

7.4.1 *How to Calculate ANOVA Manually*

For a balanced one-way layout with k treatments and n observations per treatment, write the observations as y_{ij}, where $i = 1, \ldots, k$ indexes the group and $j = 1, \ldots, n$ indexes the observation within the group. The manual workflow is:

I. Compute each group mean

$$\bar{y}_{i\bullet} = \frac{1}{n}\sum_{j=1}^{n} y_{ij}.$$

II. Compute the grand mean

$$\bar{\bar{y}} = \frac{1}{nk}\sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}.$$

III. Compute SSW/SSE by comparing observations with their own group means.

IV. Compute SSB/SSR by comparing group means with the grand mean.

V. Add SSW and SSB to get SST, then fill the ANOVA table.

Definition 7.2: Sum of Squares Within (SSW/SSE)

The **sum of squares within**, also called the **sum of squares error** (SSE), is

$$SSW = SSE = \sum_{i=1}^{k} \sum_{j=1}^{n} (y_{ij} - \bar{y}_{i\bullet})^2.$$

It measures the variation left unexplained after allowing each group to have its own mean. Equivalently, for each group:

$$\sum_{j=1}^{n} (y_{ij} - \bar{y}_{i\bullet})^2 = (n-1)s_i^2,$$

so

$$SSW = \sum_{i=1}^{k} (n-1)s_i^2$$

in a balanced design.

Definition 7.3: Sum of Squares Between (SSB/SSR)

The **sum of squares between**, also called the **sum of squares regression** (SSR), is

$$SSB = SSR = n \sum_{i=1}^{k} (\bar{y}_{i\bullet} - \bar{y})^2$$

for a balanced design. It measures the variability of the group means around the grand mean. This is the portion of total variation explained by the treatment or factor.

Definition 7.4: Total Sum of Squares (SST/TSS)

The **total sum of squares** is

$$SST = TSS = \sum_{i=1}^{k} \sum_{j=1}^{n} (y_{ij} - \bar{y})^2.$$

It is the numerator of the sample variance of all observations pooled together. The ANOVA identity

$$SST = SSB + SSW$$

splits total variation into explained between-group variation and unexplained within-group variation.

§7.5 *The F-Distribution*

The ratio of two independent chi-squared variables (divided by their d.f.) follows an F-distribution:

$$\frac{\sigma^2_{\text{between}}}{\sigma^2_{\text{within}}} \sim F_{k,n}.$$

The F-test has hypotheses:

$$H_0 : \sigma^2_{\text{between}} = \sigma^2_{\text{within}} \qquad \text{vs} \qquad H_a : \sigma^2_{\text{between}} \neq \sigma^2_{\text{within}}.$$

If sample variances are equal, $F \approx 1$.

§7.6 *ANOVA Table*

Source	**d.f.**	**SS**	**MS**	**F**
Between (treatment)	$k-1$	SSB	$\text{MSB} = \dfrac{\text{SSB}}{k-1}$	$\dfrac{\text{MSB}}{\text{MSW}}$
Within (error)	$nk-k$	SSW	$s^2 = \dfrac{\text{SSW}}{nk-k}$	
Total	$nk-1$	SST		

Table 7.1: One-Way ANOVA Table.

Compare F_{calc} to the critical value $F_{k-1,\,nk-k}$ at significance level α. The columns are filled as follows:

$$MSB = \frac{SSB}{k-1}, \qquad MSW = MSE = \frac{SSW}{nk-k},$$

$$F_{\text{calc}} = \frac{MSB}{MSW}.$$

The numerator degrees of freedom $k-1$ count how many independent group-mean contrasts are being tested. The denominator degrees of freedom $nk-k = k(n-1)$ come from within-group residual variation.

Remark Interpreting the Results

- If H_0 is *not* rejected, the factor X has no significant effect on Y.
- If H_0 *is* rejected, the factor has a significant effect, and comparing the category means reveals the nature of that effect.

§7.7 Coefficient of Determination

$$R^2 = \frac{\text{SSB}}{\text{SST}} = \frac{\text{SSB}}{\text{SSB} + \text{SSE}}.$$

R^2 represents the proportion of total variation in Y explained by the factor (treatment).

§7.8 Worked Examples

Example 7.5

Four treatments on height (inches), $n = 10$ per group:

Treatment 1	Treatment 2	Treatment 3	Treatment 4
60	50	48	47
67	52	49	67
42	43	50	54
67	67	55	67
56	67	56	68
62	59	61	65
64	67	61	65
59	64	60	56
72	63	59	60
71	65	64	65
$\bar{y}_{1\bullet} = 62.0$	$\bar{y}_{2\bullet} = 59.7$	$\bar{y}_{3\bullet} = 56.3$	$\bar{y}_{4\bullet} = 61.4$

Grand mean: $\bar{\bar{y}} = 59.85$.

Solution **Step 1 – SSB:**

$$\begin{aligned} \text{SSB} &= 10\left[(62-59.85)^2 + (59.7-59.85)^2 + (56.3-59.85)^2 + (61.4-59.85)^2\right] \\ &= 10 \times 19.65 = 196.5. \end{aligned}$$

Step 2 – SSW (within-group sum of squares):

$$\begin{aligned} \text{SSW}_1 &= (60-62)^2 + (67-62)^2 + (42-62)^2 + (67-62)^2 + (56-62)^2 \\ &\quad + (62-62)^2 + (64-62)^2 + (59-62)^2 + (72-62)^2 + (71-62)^2 = 624, \\ \text{SSW}_2 &= (50-59.7)^2 + (52-59.7)^2 + \cdots + (65-59.7)^2 = 678.1, \\ \text{SSW}_3 &= (48-56.3)^2 + (49-56.3)^2 + \cdots + (64-56.3)^2 = 332.1, \\ \text{SSW}_4 &= (47-61.4)^2 + (67-61.4)^2 + \cdots + (65-61.4)^2 = 426.4, \end{aligned}$$

$$\text{SSW} = \sum_{i=1}^{4}\sum_{j=1}^{10}(y_{ij} - \bar{y}_{i\bullet})^2 = 624 + 678.1 + 332.1 + 426.4 = 2060.6.$$

Step 3 – ANOVA Table:

Source	d.f.	SS	MS	F	p-value
Between	3	196.5	65.5	1.14	0.344
Within	36	2060.6	57.2		
Total	39	2257.1			

Conclusion: $F = 1.14 < F_{3,36,0.05}$. Fail to reject H_0. Treatment group does not significantly explain height.

$R^2 = 196.5/2257.1 \approx 9\%$: only 9% of variance in height is explained by treatment. □

Example 7.6: School Lunch Calcium Intake

Mean micronutrient intake from the school lunch across three secondary schools in England ($n = 25$ pupils each):

	School 1[a]	School 2[b]	School 3[c]
Mean Calcium (mg)	117.8	158.7	206.5
SD	62.4	70.5	86.2

[a] Most deprived; 40% subsidised lunches. [b] Medium deprived; < 10% subsidised. [c] Least deprived; private school, no subsidisation.

Grand mean: $\bar{\bar{y}} = 161$.

Solution **SSB:**

$$\text{SSB} = 25\left[(117.8-161)^2 + (158.7-161)^2 + (206.5-161)^2\right] = 98{,}113.$$

SSW:

$$\text{SSW} = 24\left[62.4^2 + 70.5^2 + 86.2^2\right] = 391{,}066.$$

ANOVA Table:

Source	d.f.	SS	MS	F	p-value
Between	2	98,113	49,056	9	< 0.05
Within	72	391,066	5,431		
Total	74	489,179			

$R^2 = 98{,}113/489{,}179 \approx 20\%$. School explains 20% of variance in calcium intake.

Conclusion: $F = 9 > F_{2,72,0.05}$. Reject H_0. Significant difference in calcium intake across schools. □

§7.9 Assumptions of One-Way ANOVA

The error term ε_{ij} must satisfy:

I. Normally distributed with zero mean.

II. Constant variance (homoscedasticity).

III. Uncorrelated with the categories of X.

IV. Mutually uncorrelated error terms.

§7.10 Beyond One-Way ANOVA

ANOVA extends naturally to more than one factor, provided the factors are independent. The total variation again partitions cleanly:

$$\mathrm{SST} = \mathrm{SSB}_1 + \mathrm{SSB}_2 + \mathrm{SSW},$$

which is the starting point for the two-way ANOVA developed in the next chapter.

8
Two-Way Analysis of Variance (ANOVA)

§8.1 Motivation

When two factors are present (e.g. treatment and a blocking variable), a **two-way ANOVA** controls for the second factor, reducing SSE and yielding more powerful tests.

§8.2 Hypotheses

Let there be k treatment groups and b blocks.

For treatments:

$$H_0 : \tau_1 = \tau_2 = \cdots = \tau_k \qquad H_a : \text{not all } \tau_i \text{ equal.}$$

For blocks:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_b \qquad H_a : \text{not all } \beta_i \text{ equal.}$$

§8.3 Partitioning of Total Variation

$$\text{SST} = \text{SSB}_1 + \text{SSB}_2 + \text{SSE},$$

where:

$$\text{SSB}_1 = b\sum_{j=1}^{k}\left(\bar{Y}_{\bullet j} - \bar{\bar{Y}}\right)^2 \quad \text{(variation among treatments),}$$

$$\text{SSB}_2 = k\sum_{i=1}^{b}\left(\bar{Y}_{i\bullet} - \bar{\bar{Y}}\right)^2 \quad \text{(variation among blocks),}$$

$$\text{SST} = \sum_{i=1}^{b}\sum_{j=1}^{k}(Y_{ij} - \bar{\bar{Y}})^2 \quad \text{(total variation),}$$

$$\text{SSE} = \text{SST} - \text{SSB}_1 - \text{SSB}_2 \quad \text{(residual / error).}$$

§8.4 *Two-Way ANOVA Table*

Source	d.f.	SS	MS	F
Treatment	$k-1$	SSB_1	$\text{MSB}_1 = \dfrac{\text{SSB}_1}{k-1}$	$\dfrac{\text{MSB}_1}{\text{MSE}}$
Block	$b-1$	SSB_2	$\text{MSB}_2 = \dfrac{\text{SSB}_2}{b-1}$	$\dfrac{\text{MSB}_2}{\text{MSE}}$
Error	$n-k-b+1$	SSE	$\text{MSE} = \dfrac{\text{SSE}}{n-k-b+1}$	
Total	$n-1$	SST		

Table 8.1: Two-Way ANOVA Table ($n = bk$ total observations).

§8.5 *F-Tests*

Test for treatment differences:

$$F = \frac{\text{MSB}_1}{\text{MSE}} = \frac{\text{SSB}_1/(k-1)}{\text{SSE}/(n-k-b+1)} \sim F_{k-1,\, n-k-b+1}.$$

Test for block differences:

$$F = \frac{\text{MSB}_2}{\text{MSE}} = \frac{\text{SSB}_2/(b-1)}{\text{SSE}/(n-k-b+1)} \sim F_{b-1,\, n-k-b+1}.$$

Reject the respective H_0 if $F_{\text{calc}} > F_{\text{critical}}$ at significance level α.

§8.6 *Mean Square Error as Pooled Variance*

MSE is analogous to the pooled variance s_p^2 from matched-samples inference:

$$\text{MSE} = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2 + \cdots + (n_k-1)s_k^2}{n-k}.$$

Theorem 8.1

MSE is an unbiased estimator of σ^2, i.e. $E(\text{MSE}) = \sigma^2$.

§8.7 Worked Example

Example 8.2

Salary data for $k = 3$ business majors (Accounting, Marketing, Finance) across $b = 6$ GPA blocks, $n = 18$ total.

	Accounting	Marketing	Finance	Block mean
Block 1	41	45	51	45.67
Block 2	36	38	45	39.67
Block 3	27	33	31	30.83
Block 4	32	29	35	32.00
Block 5	26	31	32	29.67
Block 6	23	25	27	25.00
Trt. mean	30.83	33.50	36.83	

Grand mean: $\bar{\bar{Y}} = 33.72$.

Solution **SSB_1 (treatments):**

$$\text{SSB}_1 = 6\left[(30.83 - 33.72)^2 + (33.5 - 33.72)^2 + (36.83 - 33.72)^2\right] = 108.4.$$

SSB_2 (blocks):

$$\text{SSB}_2 = 3\left[(45.67 - 33.72)^2 + (39.67 - 33.72)^2 + \cdots + (25 - 33.72)^2\right] = 854.9.$$

SST:

$$\text{SST} = \sum_{i,j}(Y_{ij} - \bar{\bar{Y}})^2 = 1015.61.$$

SSE:

$$\text{SSE} = 1015.61 - 108.4 - 854.9 = 52.2, \qquad \text{d.f.} = 18 - 3 - 6 + 1 = 10.$$

F-test for treatments:

$$F = \frac{108.4/2}{52.2/10} = \frac{54.2}{5.2} = 10.4 \quad > \quad F_{0.05,2,10} = 4.10.$$

Reject H_0: significant salary differences across majors.

F-test for blocks:

$$F = \frac{854.9/5}{52.2/10} = \frac{170.98}{5.2} = 32.76 \quad > \quad F_{0.05,5,10} = 3.33.$$

Reject H_0: significant salary differences across GPA blocks. □

9
Maximum Likelihood Estimation

§9.1 Likelihood and Maximum Likelihood

Definition 9.1: Likelihood

Suppose an experiment or sample produces an observed event or data vector E, and suppose the probability model depends on a parameter θ. The **likelihood** of θ is

$$\mathcal{L}(E;\theta) = P(E;\theta)$$

for discrete data, or the joint density evaluated at the observed data for continuous data. The semicolon means that θ is part of the probability rule, not an event being conditioned on.

Definition 9.2: Maximum Likelihood Estimator

The maximum likelihood estimator (MLE) is the parameter value that maximises the likelihood:

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \mathcal{L}(E;\theta).$$

Because $\log(x)$ is increasing, the maximiser of $\mathcal{L}$ is also the maximiser of the log-likelihood

$$\ell(\theta) = \log \mathcal{L}(E;\theta).$$

§9.2 *Coin-Flip Example*

Example 9.3: Coin Flips

A coin is flipped independently 10 times and the observed sequence is

$$H\,T\,T\,T\,H\,H\,T\,H\,H\,H.$$

Estimate the probability θ that the coin comes up heads.

Solution The sequence contains 6 heads and 4 tails, so the likelihood is

$$\mathcal{L}(\theta) = \theta^6(1-\theta)^4, \qquad 0 \le \theta \le 1.$$

The direct derivative is

$$\frac{d}{d\theta}\theta^6(1-\theta)^4 = 6\theta^5(1-\theta)^4 - 4\theta^6(1-\theta)^3.$$

Setting this to zero and ignoring endpoint factors gives

$$6(1-\hat{\theta}) = 4\hat{\theta} \quad \Longrightarrow \quad \hat{\theta} = \frac{6}{10} = \frac{3}{5}.$$

The endpoints $\theta = 0$ and $\theta = 1$ give likelihood zero, while $\theta = 3/5$ gives a positive likelihood; hence the MLE is $\hat{\theta} = 3/5$.

Using the log-likelihood is simpler:

$$\ell(\theta) = 6\log\theta + 4\log(1-\theta).$$

$$\ell'(\theta) = \frac{6}{\theta} - \frac{4}{1-\theta}.$$

Solving $\ell'(\theta) = 0$ gives the same $\hat{\theta} = 3/5$. The second derivative

$$\ell''(\theta) = -\frac{6}{\theta^2} - \frac{4}{(1-\theta)^2} < 0$$

confirms the critical point is a maximum. □

§9.3 *General MLE Workflow*

For n independent observations $x_1, \ldots, x_n$:

I. Write the likelihood. For discrete data,

$$\mathcal{L}(\theta) = \prod_{i=1}^{n} P(X = x_i; \theta).$$

For continuous data,

$$\mathcal{L}(\theta) = \prod_{i=1}^{n} f_X(x_i; \theta).$$

II. Take logs to obtain $\ell(\theta) = \log \mathcal{L}(\theta)$.

III. Differentiate, set the derivative or score equal to zero, and solve.

IV. Check endpoints or use the second derivative/Hessian to confirm the maximum.

§9.4 *Normal Mean with Known Variance*

Example 9.4: Normal Mean, Variance Known

Let $x_1, \dots, x_n$ be independent draws from $N(\mu, \sigma^2)$, where μ is unknown and σ^2 is known. Find the MLE of μ.

Solution The likelihood is

$$\mathcal{L}(\mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x_i - \mu)^2}{2\sigma^2}\right].$$

The log-likelihood is

$$\ell(\mu) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2.$$

Differentiate:

$$\ell'(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i - \mu).$$

Set equal to zero:

$$\sum_{i=1}^{n} (x_i - \hat{\mu}) = 0 \quad \Longrightarrow \quad \sum_{i=1}^{n} x_i = n\hat{\mu} \quad \Longrightarrow \quad \hat{\mu} = \bar{x}.$$

Since

$$\ell''(\mu) = -\frac{n}{\sigma^2} < 0,$$

the sample mean is the maximiser. □

§9.5 *Normal Mean and Variance Both Unknown*

Theorem 9.5: Normal MLEs

If $x_1, \dots, x_n$ are independent observations from $N(\mu, \sigma^2)$ with both μ and σ^2 unknown, then

$$\hat{\mu}_{\text{MLE}} = \bar{x}, \qquad \hat{\sigma}^2_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

Proof The likelihood is

$$\mathcal{L}(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x_i - \mu)^2}{2\sigma^2}\right].$$

The log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Differentiate with respect to μ:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu).$$

Setting this equal to zero gives $\hat{\mu} = \bar{x}$.

Let $v = \sigma^2$. Differentiate with respect to v:

$$\frac{\partial \ell}{\partial v} = -\frac{n}{2v} + \frac{1}{2v^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Set to zero and multiply by $2v^2$:

$$-nv + \sum_{i=1}^{n}(x_i - \mu)^2 = 0 \quad \Longrightarrow \quad \hat{v} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Substituting $\hat{\mu} = \bar{x}$ gives

$$\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

The MLE uses divisor n, not $n-1$; it maximises likelihood but is biased for σ^2. □

§9.6 Standard Distribution MLEs

Example 9.6: Poisson MLE

Let $y_1, \ldots, y_n$ be independent observations from a Poisson distribution with parameter $\lambda > 0$:

$$f(y_i; \lambda) = \frac{\lambda^{y_i} e^{-\lambda}}{y_i!}, \qquad y_i = 0, 1, 2, \ldots.$$

Find the MLE of λ.

Solution

$$\mathcal{L}(\lambda) = \prod_{i=1}^{n}\frac{\lambda^{y_i} e^{-\lambda}}{y_i!} = \frac{\lambda^{\sum y_i} e^{-n\lambda}}{\prod y_i!}.$$

$$\ell(\lambda) = \left(\sum y_i\right)\log\lambda - n\lambda - \sum\log(y_i!).$$

$$\ell'(\lambda) = \frac{\sum y_i}{\lambda} - n.$$

Set to zero:

$$\hat{\lambda} = \frac{1}{n}\sum_{i=1}^{n} y_i = \bar{y}.$$

Also $\ell''(\lambda) = -(\sum y_i)/\lambda^2 < 0$ when the sample contains at least one event, so this is a maximum. □

Example 9.7: Binomial MLE

Let $y_1, \ldots, y_m$ be independent observations from Binomial(n, p), where n is known and $0 < p < 1$:

$$f(y_i; p) = \binom{n}{y_i} p^{y_i} (1-p)^{n-y_i}.$$

Find the MLE of p.

Solution

$$\ell(p) = \sum_{i=1}^{m} \log \binom{n}{y_i} + \left(\sum y_i\right) \log p + \left(mn - \sum y_i\right) \log(1-p).$$

$$\ell'(p) = \frac{\sum y_i}{p} - \frac{mn - \sum y_i}{1-p}.$$

Set to zero:

$$(1-\hat{p}) \sum y_i = \hat{p}\left(mn - \sum y_i\right) \quad \Longrightarrow \quad \hat{p} = \frac{\sum_{i=1}^{m} y_i}{mn}.$$

Thus the MLE is total successes divided by total trials. □

Example 9.8: Exponential MLE

Let $y_1, \ldots, y_n$ be independent observations from an exponential distribution with rate parameter $\theta > 0$:

$$f(y_i; \theta) = \theta e^{-\theta y_i}, \qquad y_i \geq 0.$$

Find the MLE of θ.

Solution

$$\mathcal{L}(\theta) = \prod_{i=1}^{n} \theta e^{-\theta y_i} = \theta^n \exp\left(-\theta \sum y_i\right).$$

$$\ell(\theta) = n \log \theta - \theta \sum y_i.$$

$$\ell'(\theta) = \frac{n}{\theta} - \sum y_i.$$

Set to zero:

$$\hat{\theta} = \frac{n}{\sum y_i} = \frac{1}{\bar{y}}.$$

Because $\ell''(\theta) = -n/\theta^2 < 0$, this is the maximum. □

Example 9.9: Normal Variance, Mean Known

Let $y_1, \ldots, y_n$ be independent observations from $N(\mu, \sigma^2)$, where μ is known and σ^2 is unknown. Find the MLE of σ^2.

Solution Let $v = \sigma^2$. The log-likelihood, up to constants, is

$$\ell(v) = -\frac{n}{2} \log v - \frac{1}{2v} \sum_{i=1}^{n} (y_i - \mu)^2.$$

$$\ell'(v) = -\frac{n}{2v} + \frac{1}{2v^2} \sum_{i=1}^{n} (y_i - \mu)^2.$$

Setting $\ell'(v) = 0$ gives

$$\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \mu)^2.$$

$\square$

10

Fisher Information

§10.1 Definition

Definition 10.1: Fisher Information

For independent observations with log-likelihood $\ell(\theta)$, the Fisher information for a scalar parameter θ is

$$I(\theta) = -E\left[\frac{d^2}{d\theta^2}\ell(\theta)\right].$$

For a realised sample, the observed information is

$$J(\theta) = -\frac{d^2}{d\theta^2}\ell(\theta).$$

If the sample has n independent observations, information usually adds: $I_n(\theta) = nI_1(\theta)$.

§10.2 Fisher Information for Common Models

Example 10.2: Poisson Fisher Information

For $Y_1, \ldots, Y_n \overset{iid}{\sim} \text{Poisson}(\lambda)$, find $I(\lambda)$.

Solution

$$\ell(\lambda) = \left(\sum y_i\right)\log\lambda - n\lambda - \sum \log(y_i!).$$

$$\ell'(\lambda) = \frac{\sum y_i}{\lambda} - n, \qquad \ell''(\lambda) = -\frac{\sum y_i}{\lambda^2}.$$

Therefore

$$I(\lambda) = -E[\ell''(\lambda)] = E\left[\frac{\sum Y_i}{\lambda^2}\right] = \frac{n\lambda}{\lambda^2} = \frac{n}{\lambda}.$$

□

Example 10.3: Exponential Fisher Information

For $Y_1, \dots, Y_n \overset{iid}{\sim} \text{Exponential}(\theta)$ with rate θ, find $I(\theta)$.

Solution

$$\ell(\theta) = n \log \theta - \theta \sum y_i.$$

$$\ell'(\theta) = \frac{n}{\theta} - \sum y_i, \qquad \ell''(\theta) = -\frac{n}{\theta^2}.$$

Thus

$$I(\theta) = -E[\ell''(\theta)] = \frac{n}{\theta^2}.$$

□

Example 10.4: Normal Mean Fisher Information

For $Y_1, \dots, Y_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with known σ^2, find the Fisher information for μ.

Solution

$$\ell(\mu) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (y_i - \mu)^2.$$

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum (y_i - \mu), \qquad \frac{\partial^2 \ell}{\partial \mu^2} = -\frac{n}{\sigma^2}.$$

Therefore

$$I(\mu) = \frac{n}{\sigma^2}.$$

□

Example 10.5: Normal Variance Fisher Information

For $Y_1, \dots, Y_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with known μ, find the Fisher information for $v = \sigma^2$.

Solution

$$\ell(v) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log v - \frac{1}{2v} \sum (y_i - \mu)^2.$$

$$\frac{\partial \ell}{\partial v} = -\frac{n}{2v} + \frac{1}{2v^2} \sum (y_i - \mu)^2,$$

$$\frac{\partial^2 \ell}{\partial v^2} = \frac{n}{2v^2} - \frac{1}{v^3} \sum (y_i - \mu)^2.$$

Since $E[\sum (Y_i - \mu)^2] = nv$,

$$I(v) = -E\left[\frac{\partial^2 \ell}{\partial v^2}\right] = -\left(\frac{n}{2v^2} - \frac{nv}{v^3}\right) = \frac{n}{2v^2}.$$

Thus

$$I(\sigma^2) = \frac{n}{2\sigma^4}.$$

□

Example 10.6: Normal Fisher Information Matrix

For $Y_1, \ldots, Y_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with both parameters unknown, find the Fisher information matrix for (μ, v) where $v = \sigma^2$.

Solution Because $v = \sigma^2$, the log-likelihood is

$$\ell(\mu, v) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log v - \frac{1}{2v}\sum_{i=1}^{n}(y_i - \mu)^2.$$

The score components are

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{v}\sum_{i=1}^{n}(y_i - \mu), \qquad \frac{\partial \ell}{\partial v} = -\frac{n}{2v} + \frac{1}{2v^2}\sum_{i=1}^{n}(y_i - \mu)^2.$$

The second derivatives are

$$\frac{\partial^2 \ell}{\partial \mu^2} = -\frac{n}{v}, \qquad \frac{\partial^2 \ell}{\partial \mu\, \partial v} = -\frac{1}{v^2}\sum_{i=1}^{n}(y_i - \mu),$$

and

$$\frac{\partial^2 \ell}{\partial v^2} = \frac{n}{2v^2} - \frac{1}{v^3}\sum_{i=1}^{n}(y_i - \mu)^2.$$

The Fisher information is the negative expected Hessian. Since

$$E\left[\sum_{i=1}^{n}(Y_i - \mu)\right] = 0, \qquad E\left[\sum_{i=1}^{n}(Y_i - \mu)^2\right] = nv,$$

the entries are

$$I_{\mu\mu} = -E\left(\frac{\partial^2 \ell}{\partial \mu^2}\right) = \frac{n}{v},$$

$$I_{\mu v} = I_{v\mu} = -E\left(\frac{\partial^2 \ell}{\partial \mu\, \partial v}\right) = \frac{1}{v^2}E\left[\sum_{i=1}^{n}(Y_i - \mu)\right] = 0,$$

and

$$I_{vv} = -E\left(\frac{\partial^2 \ell}{\partial v^2}\right) = -\left(\frac{n}{2v^2} - \frac{nv}{v^3}\right) = \frac{n}{2v^2}.$$

Hence

$$I(\mu, v) = \begin{pmatrix} n/v & 0 \\ 0 & n/(2v^2) \end{pmatrix}.$$

Equivalently, in terms of σ^2,

$$I(\mu, \sigma^2) = \begin{pmatrix} n/\sigma^2 & 0 \\ 0 & n/(2\sigma^4) \end{pmatrix}.$$

□

11
Summary of Key Formulae

§11.1 One-Sample Inference for Mean μ (Weeks 1–2, Lec. 1)

Large-sample CI: $\bar{y} \pm z_{\alpha/2} \dfrac{\sigma}{\sqrt{n}}$

Small-sample CI: $\bar{y} \pm t_{\alpha/2,n-1} \dfrac{s}{\sqrt{n}}$

Large-sample Z-test: $Z_{\text{obs}} = \dfrac{\bar{y} - \mu_0}{\sigma/\sqrt{n}}$

Small-sample t-test: $t_{\text{obs}} = \dfrac{\bar{y} - \mu_0}{s/\sqrt{n}}, \quad \nu = n - 1$

Power (upper-tail): $P(\bar{Y} \geq c \mid \mu = \mu_a), \quad c = \mu_0 + z_\alpha \dfrac{\sigma}{\sqrt{n}}$

§11.2 Two-Sample Inference for $\mu_1 - \mu_2$ (Week 2, Lec. 2)

Large-sample CI: $(\bar{y}_1 - \bar{y}_2) \pm z_{\alpha/2}\sqrt{S_1^2/n_1 + S_2^2/n_2}$

Pooled variance: $S_p^2 = \dfrac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$

Pooled t-statistic: $t_{\text{obs}} = \dfrac{(\bar{y}_1 - \bar{y}_2) - \Delta_0}{\sqrt{S_p^2(1/n_1 + 1/n_2)}}, \quad \nu = n_1 + n_2 - 2$

Paired t-statistic: $t_{\text{obs}} = \dfrac{\bar{d} - \Delta_0}{s_d/\sqrt{n}}, \quad \nu = n - 1$

§11.3 One-Sample Inference for Proportion p (Week 3)

$$\hat{p} = \frac{X}{n}, \qquad \text{CI: } \hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \qquad Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$$

§11.4 Two-Sample Inference for $p_1 - p_2$ (Week 3)

$$\text{CI: } (\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}, \qquad Z_{\text{obs}} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}}$$

§11.5 Simple and Multiple Linear Regression (Week 5)

Simple Linear Model: $Y = \beta_0 + \beta_1 X + \varepsilon$

Least Squares Slope: $\hat{\beta}_1 = \dfrac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$

Least Squares Intercept: $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

Multiple Linear Model: $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k + \varepsilon$

Matrix Estimation: $\widehat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$

Sample Correlation: $r = \dfrac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$

Coefficient of Determination: $R^2 = \dfrac{SSR}{SST} = 1 - \dfrac{SSE}{SST}, \qquad R^2 = r^2$ (simple)

§11.6 Polynomial Regression (Week 6)

Polynomial Model: $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_{N-1} X^{N-1}$

Simple Slope Formula: $\dfrac{\partial Y}{\partial X} = \beta_1 + 2\beta_2 X + 3\beta_3 X^2 + \cdots$

Testing Higher-Order Terms: Compare augmented vs. compact via F- or t-test.

§11.7 Simple Linear Regression (Week 9)

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}, \quad S_e^2 = \frac{S_{yy} - \hat{\beta}_1 S_{xy}}{n-2}.$$

$$\text{Var}(\hat{\beta}_0) = \sigma^2\left(\tfrac{1}{n} + \tfrac{\bar{X}^2}{S_{xx}}\right), \quad \text{Var}(\hat{\beta}_1) = \tfrac{\sigma^2}{S_{xx}}, \quad \text{Var}(\hat{e}_i) = \sigma^2\left(1 - \tfrac{1}{n} - \tfrac{(X_i - \bar{X})^2}{S_{xx}}\right).$$

§11.8 One-Way ANOVA (Week 10)

$$\text{SST} = \text{SSB} + \text{SSW}, \quad F = \frac{\text{SSB}/(k-1)}{\text{SSW}/(nk-k)}, \quad R^2 = \frac{\text{SSB}}{\text{SST}}.$$

§11.9 Two-Way ANOVA (Week 11)

$$\mathrm{SST} = \mathrm{SSB}_1 + \mathrm{SSB}_2 + \mathrm{SSE}.$$

$$F_{\text{treatment}} = \frac{\mathrm{SSB}_1/(k-1)}{\mathrm{SSE}/(n-k-b+1)}, \quad F_{\text{block}} = \frac{\mathrm{SSB}_2/(b-1)}{\mathrm{SSE}/(n-k-b+1)}.$$